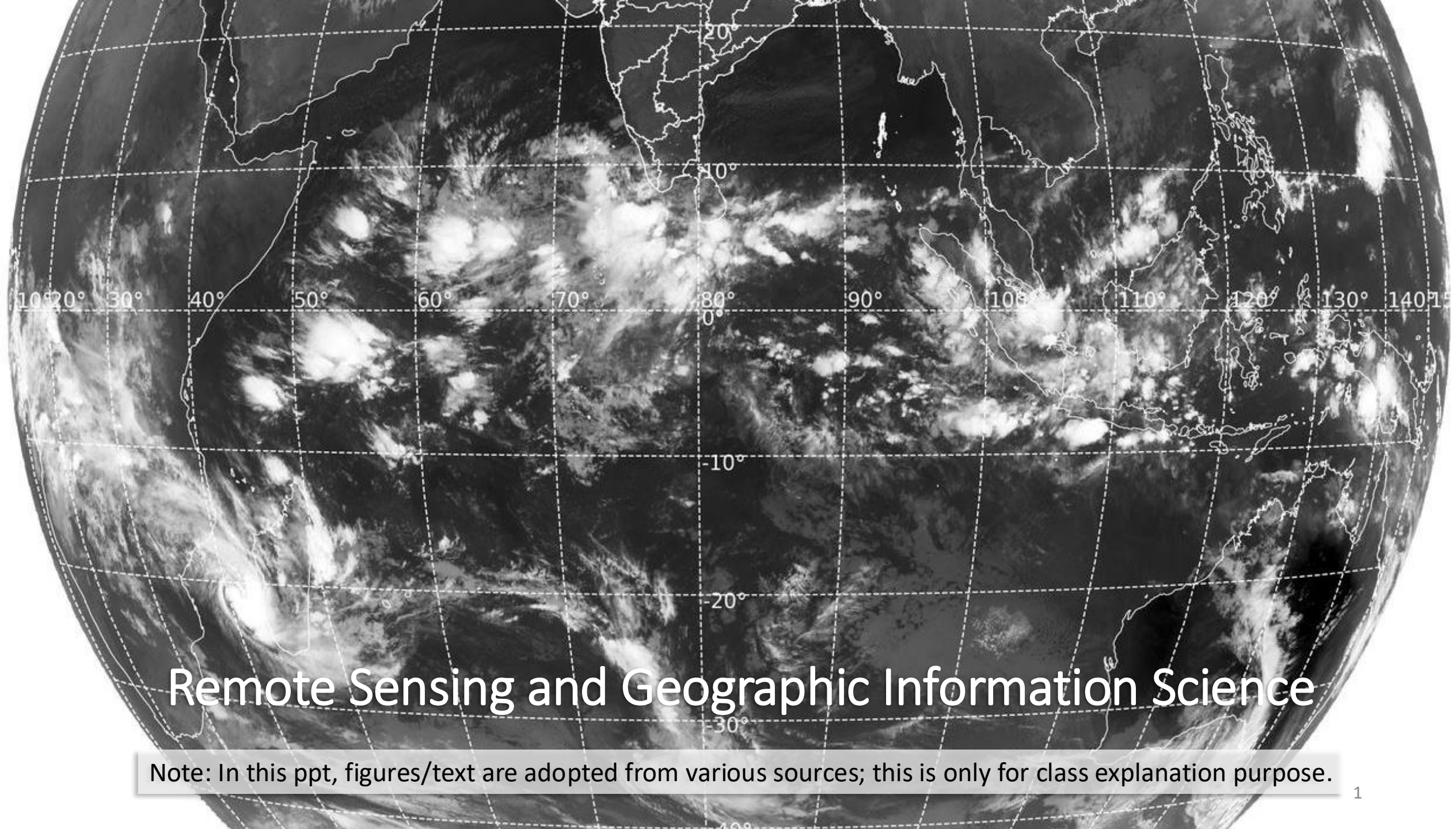




Remote Sensing and Geographic Information Science

Note: In this ppt, figures/text are adopted from various sources; this is only for class explanation purpose.

What is a Geographic Data?

Geographic data are built up from facts about the geographic world. At its most primitive, an atom of geographic data links a **place**, often a **time**, with **some property** of that place (and time).

The first of these, place, is specified in one of several ways (Place name, postal code, lat/long etc.) and there are also many ways of specifying the second, time. We often use the term ***attribute*** to refer to the last of these three.

Geographic data link place, time, and attributes.

For example, consider the statement 'The temperature at local noon on December 2nd 2004 at latitude 43 degrees 45 minutes north, longitude 78 degrees 0 minutes west, was 8 degrees Celsius'. It ties location and time to the property or attribute of atmospheric temperature.

Geographical Data Types

- **Spatial data**

Gives information about location (*where it is ?*) and spatial dimensions of geographical entities.

- **Non-spatial data**

Gives information about the properties of a feature (*What is it ?*)

- Attributes can be physical, social, economic, demographic, environmental, etc.

Geographical (Spatial) Data Representation

The two fundamental ways of representing geography are discrete objects and continuous fields.

Discrete Objects

We prefer a world of black and white, of good guys and bad guys, to the real world of shades of gray.

This preference is reflected in one way of viewing the geographic world, known as the *discrete object* view.

In this view, the world is empty, except where it is occupied by objects with **well-defined boundaries** that are instances of generally recognized categories.

Countable

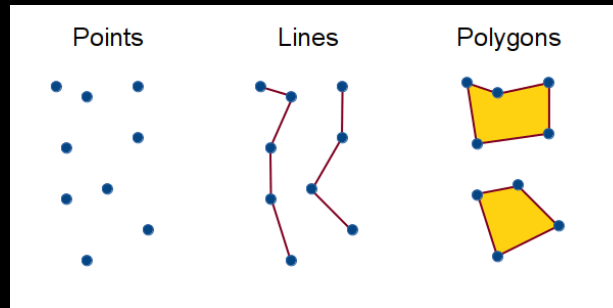
Geographical (Spatial) Data Representation- Discrete objects Model

Spatial objects in the real world can be thought of as occurring as three easily identifiable types:

Points

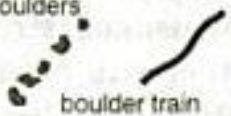
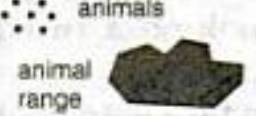
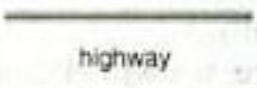
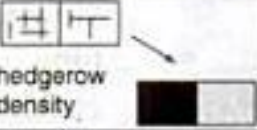
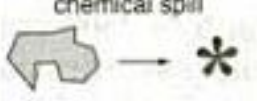
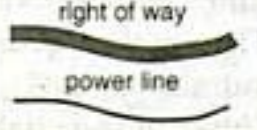
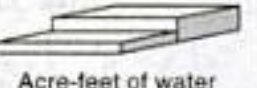
Lines

Areas/Surfaces



Collectively, these three types can represent most of the tangible natural and human phenomena that we encounter on an everyday basis.

Comparison of Real –World Phenomena and the Cartographer's Conception

Cartographer's Conception					
		point representation	line representation	area representation	volumetric representation
Real World Phenomena	point objects	 tree			
	line objects	 airport	 highway		
	area objects	 chemical spill		 new subdivision	
	volumetric objects	 Open-pit mine		 irrigation drain	 Acre-feet of water

Source: Michael N. Demers, 2008

Point, line, area and surface features with examples

Geographical (Spatial) Data Representation- Continuous Model

While we might think of terrain as composed of discrete mountain peaks, valleys, ridges, slopes, etc., and think of listing them in tables and counting them, there are unresolvable problems of definition for all of these objects. Instead, it is much more useful to think of terrain as a continuous surface, in which elevation can be defined rigorously at every point. Such continuous surfaces form the basis of the other common view of geographic phenomena, known as the *continuous field* view (and not to be confused with other meanings of the word field).

In this view, the geographic world can be described by a number of *variables*, each measurable at any point on the Earth's surface, and changing in value across the surface.

Examples of Fields

- Soil properties, e.g. pH, soil moisture
- Population density
 - But at fine enough scale the concept breaks down
- Identity of land owner
 - A single value of a nominal property at any point
- Name of county or state or nation
- Atmospheric temperature, pressure

Object based view vs Continuous view

Objects are distinguished by their dimensions, and naturally fall into categories of points, lines, or areas.

Continuous fields, on the other hand, can be distinguished by what varies, and how smoothly.

Data (Non-Spatial/*a*spatial) Representation

One can indicate that particular feature, with a particular name and with some measurable properties, exists in a documented location. But before we assign these properties and attributes, we must know how to measure them. Otherwise we cannot compare the objects at one location with those at another.

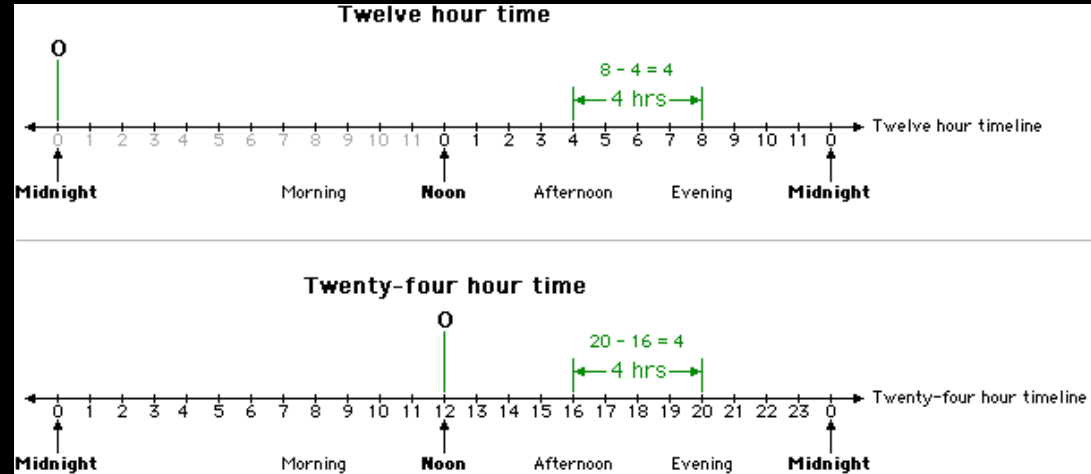
- What meter stick do we use?
- How accurately can we describe the objects we encounter?
- What effects will these different levels of measurement have on our ability to compare them?

Scale/Levels of Geographic Data Measurements

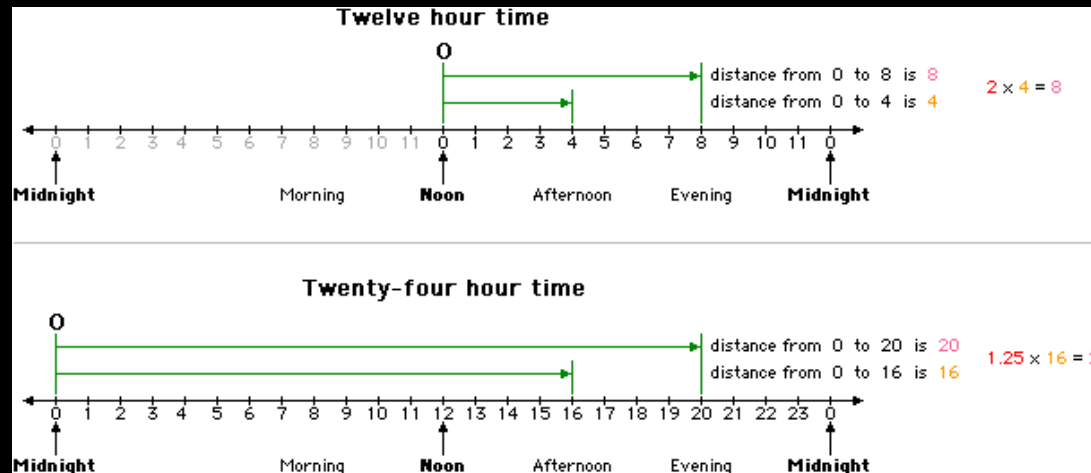
There is well defined measurement framework for nearly all forms of data, including geographic data. Any data in GIS can be represented by one of the following kinds of scales.

- **Nominal**
 - Names or labels, if numbers are used the values have no mathematical relationship.
- **Ordinal**
 - Attributes are ordered in sequence, there is no mathematical relationship but we can make comparison of size.
- **Interval**
 - has a measured numerical value as an associated attribute. The **difference** (interval) between numbers is meaningful mathematically, but the numbering scale does not start at 0. **Interval between the values is interpretable.** *Ratios of differences* can be expressed; for example, one difference can be twice another.
- **Ratio**
 - is a measured value on a scale with a meaningful zero on which absolute mathematical operations can be performed.

Illustrative Example for Interval Data



**Time Measured in
12-hr clock and
24- hr clock**



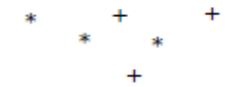
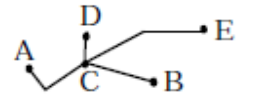
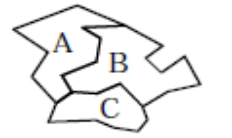
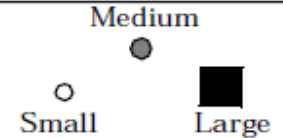
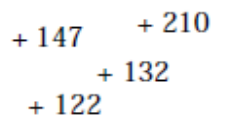
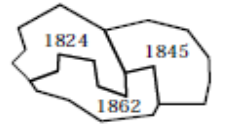
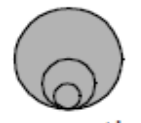
If we look at the ratios of these numbers [$8/4 = 2$, and $20/16 = 1.125$], the ratios are different, indicating that these ratios have no meaning.

Why is Level of Measurement Important?

First, knowing the level of measurement helps you decide **how to interpret the data from that variable**. When you know that a measure is nominal, then you know that the numerical values are just short codes for the longer names.

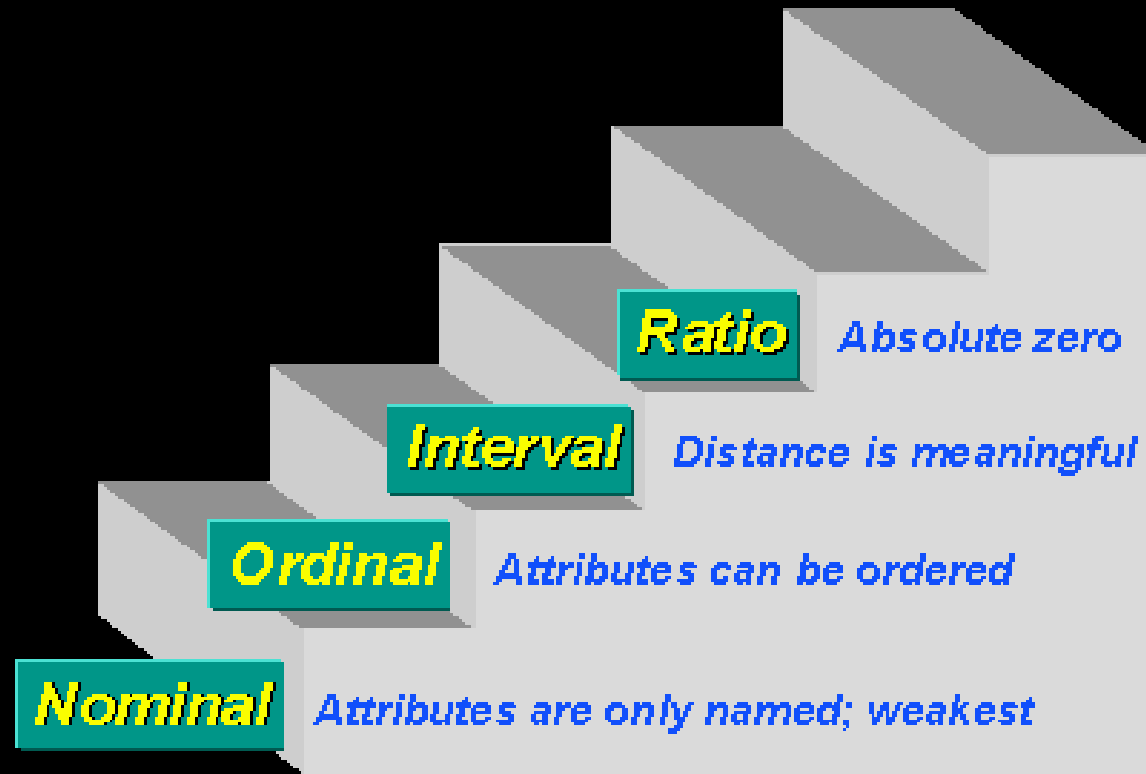
Second, knowing the level of measurement helps you decide **what statistical analysis is appropriate on the values that were assigned**. If a measure is nominal, then you know that you would never average the data values or do a t-test on the data.

Measurement Levels of Cartographic Objects

Content scaling level	Defining relations	FORM OF CARTOGRAPHIC SYMBOL		
		POINT	LINE	AREA
Nominal	Equivalence	 Wholesale and retail establishments	 Highway connectivity	 Land ownership
Ordinal	Equivalence Greater than	 Population centers	 Roads by degree of improvement	 Crop yield
Interval	Equivalence Greater than Ratio of intervals	 Spot elevations	 Graticule	 Date of settlement
Ratio	Equivalence Greater than Ratio of intervals Ratio of scale values	 Area proportional to population	 Population density isopleths	 Value proportional to population density

Robinson and Sale, *Elements of Cartography*, 3d ed. , 1969, by John Wiley & Sons

Four Levels of Measurement



Summary of Data Types and Scale Measures

Provides:	Nominal	Ordinal	Interval	Ratio
"Counts," aka "Frequency of Distribution"	✓	✓	✓	✓
Mode, Median		✓	✓	✓
The "order" of values is known		✓	✓	✓
Can quantify the difference between each value			✓	✓
Can add or subtract values			✓	✓
Can multiple and divide values				✓
Has "true zero"				✓

Attribute Data Scales: Key Points

Nominal

	Nigeria		Russia
	Poland		China
	U.K.		India
	France		Turkey
	U.S.A.		Brazil
	Australia		Argentina
	Spain		Mexico
	Israel		Jamaica

- **Categories/Labels**
- No meaningful rank order

Ordinal



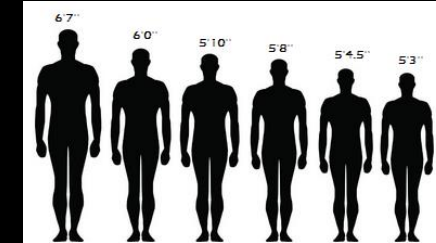
- **Ranks**
- Have a meaningful order
- Inconsistent intervals between consecutive values

Interval



- **Consistent difference** between consecutive values
- Arbitrary zero
- Difference measurement only

Ratio



- Consistent difference between consecutive values
- **Well defined zero**
- Offer the greatest flexibility for statistical analysis

CATEGORICAL DATA

Do not provide for reasonable numerical measurements

QUANTITATIVE DATA

Facilitate numerical measurements

Spatial Data Models/Structures

- **Vector Models**— based on points, lines and Polygons
- **Raster Models** — based on grids/pixels

PoP Quiz-III

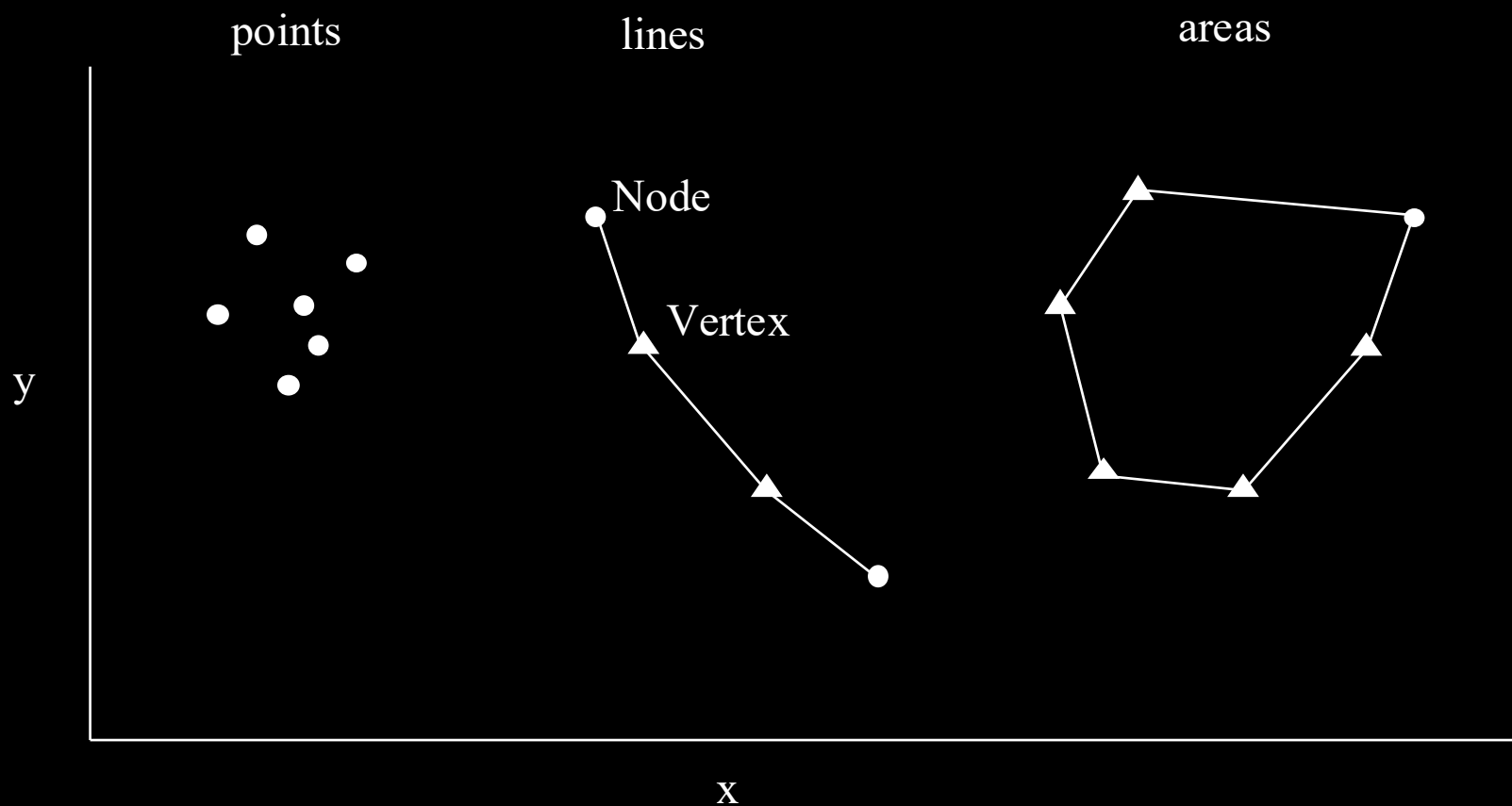
1. What is the difference between raster and vector dataset?
2. What is attribute data in GIS?
3. What is the difference between Nominal and Ordinal dataset?

Vector Data Model

What is Vector Data Model?

- **Geographers typically model the world with objects located at different places on the surface of the earth.**
- **Through this model, real -world entities are represented simply by:**
 - **points**
 - **lines**
 - **polygons**
- **These “shapes” represent the spatial character of the object in terms of shape, size, spatial arrangement.**

Vector Structures of Various Features



Varieties of Vector Data Formats

- **Triangulated irregular network (TIN)**
- **DIME files and TIGER files**
- **Network model**
- **Digital Line Graph (DLG)**
- **Shapefile (ArcView/ArcGIS; ESRI)**
- **Others: HPGL, PostScript/ASCII, CAD/.dxf**

Vector Data Models

- **Spaghetti** vector data model
- **Topologic** vector data model

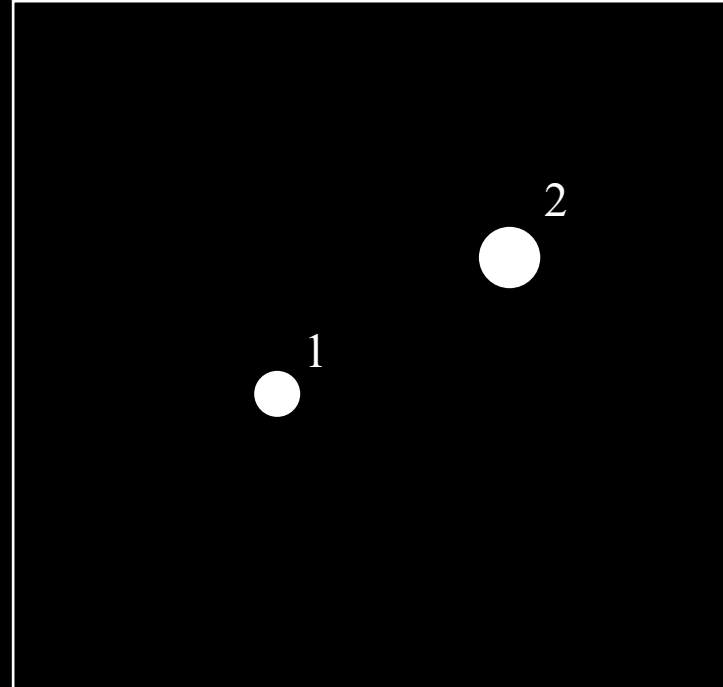
Spaghetti Vector Data Model

- *Spaghetti* data structure represents a planar configuration of points, arcs, and areas.

Each point, line, or polygon is stored as a record in a file consisting →

- Entity ID, and
- List of coordinates that define geometry

For Points:



ID Coordinates	
1	3,4
2	5,5

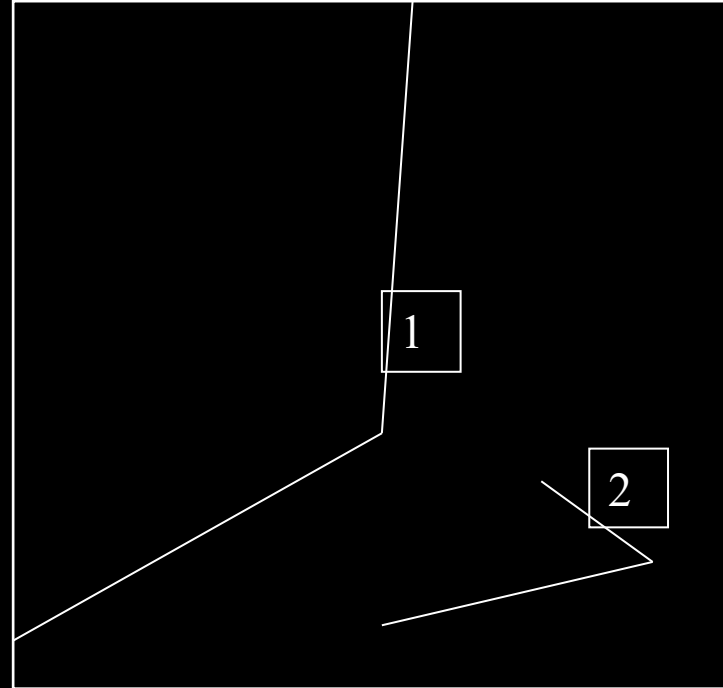
Spaghetti Vector Data Model

- *Spaghetti* data structure represents a planar configuration of points, arcs, and areas.

Each point, line, or polygon is stored as a record in a file consisting →

- Entity ID, and
- List of coordinates that define geometry

For Lines:



ID Coordinates

1	(0,1), (3,4), (5,6)
2	(3,1), (5,2), (4,3)

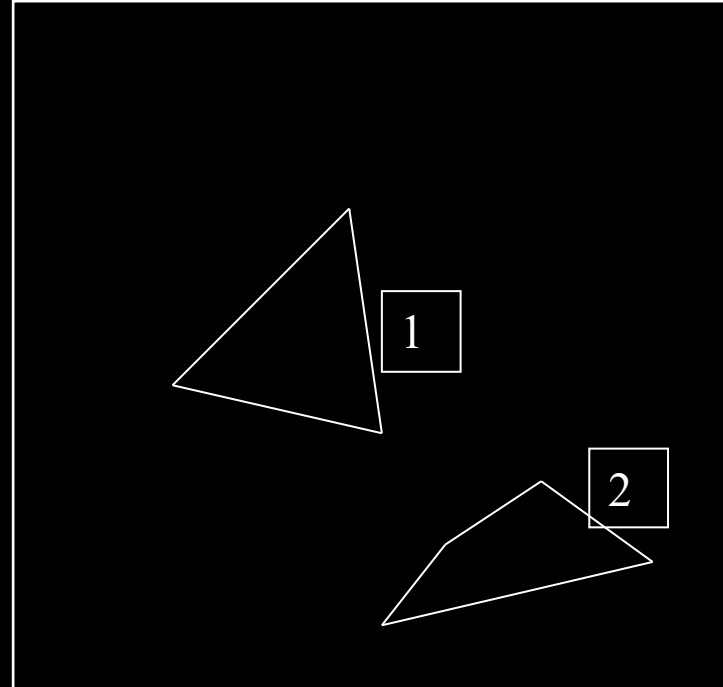
Spaghetti Vector Data Model

- *Spaghetti* data structure represents a planar configuration of points, arcs, and areas.

Each point, line, or polygon is stored as a record in a file consisting →

- Entity ID, and
- List of coordinates that define geometry

For Polygons:



ID Coordinates

1 (2,4), (4,3), (3,6) , (2,4),

2 (3,1), (5,2), (4,3), (3,2), (3,1)

Spaghetti Vector Data Model

Advantages

- simple

- efficient for display and plotting

Disadvantages

- inefficient for most types of spatial analysis

Spaghetti Vector Format

There is **NO** explicit representation of **topological inter-relationships** among the entities

So What is Topology?....

- A set of rules on how objects relate to each other.
- Consider these relationships:

adjacency

containment

connectivity

Example:

- which polygons are next to which? (e.g., Which districts are adjacent to each other?)
- which features are within another feature (e.g., “island polygons” -- an island within a lake)
- which lines connect to which? (e.g., which roads do I take to get from here to the hospital?)

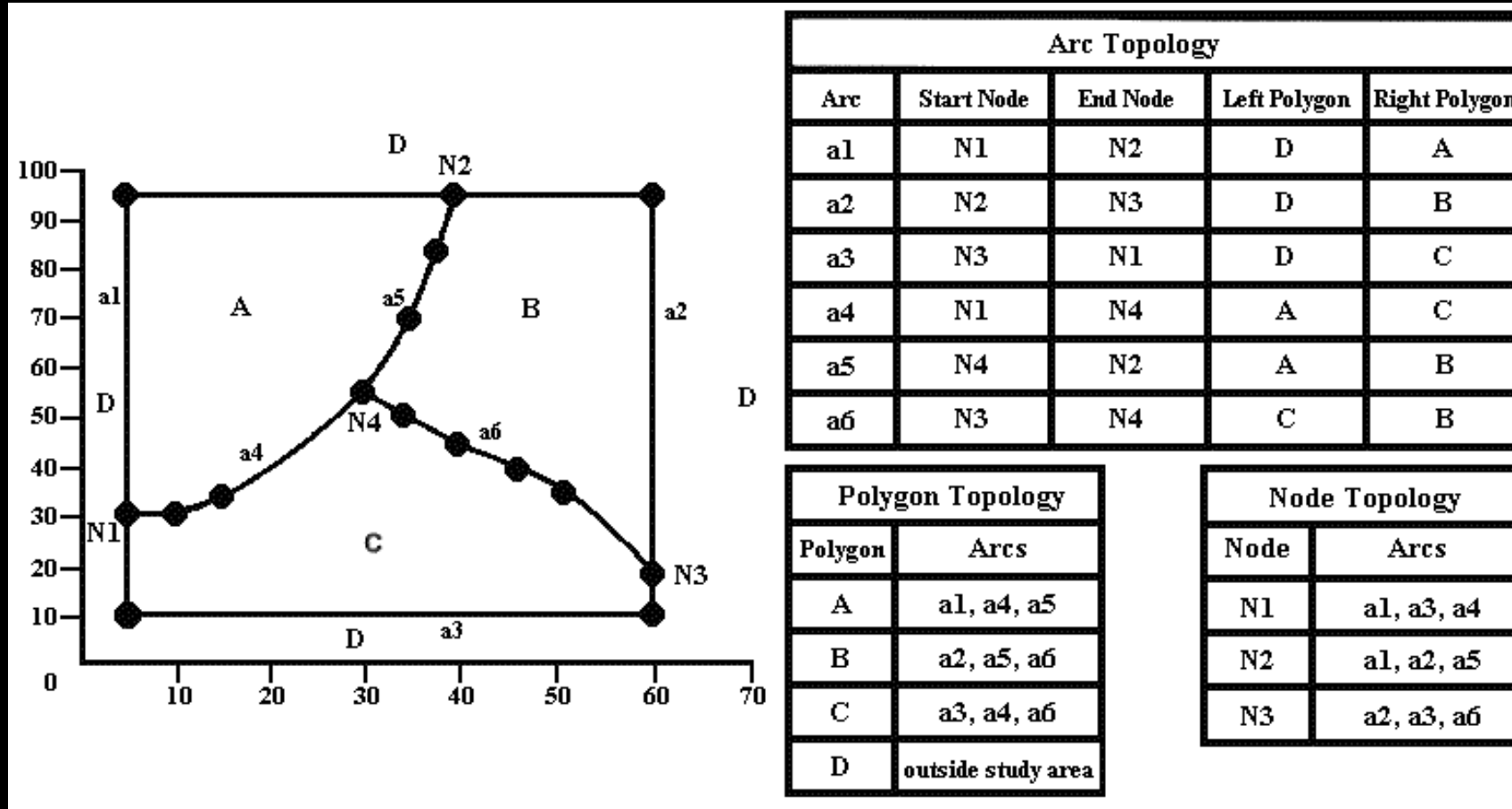
Topologic Vector Data Model

Composed of points, lines, and polygons

In addition to coordinate locations, the topologic relationships among geometric features are explicitly recorded.

Different topological relations exists for specifically node, arc and polygons.

Topologic Vector Data Model



Source: Greg Reed, 2010

http://library.oceanteacher.org/OTMediawiki/index.php/Vector_Data_Topology

Advantages of Topologic Vector Data Model

Data is stored efficiently, so large data sets can be processed quickly.

Facilitates analytical functions, e.g.,

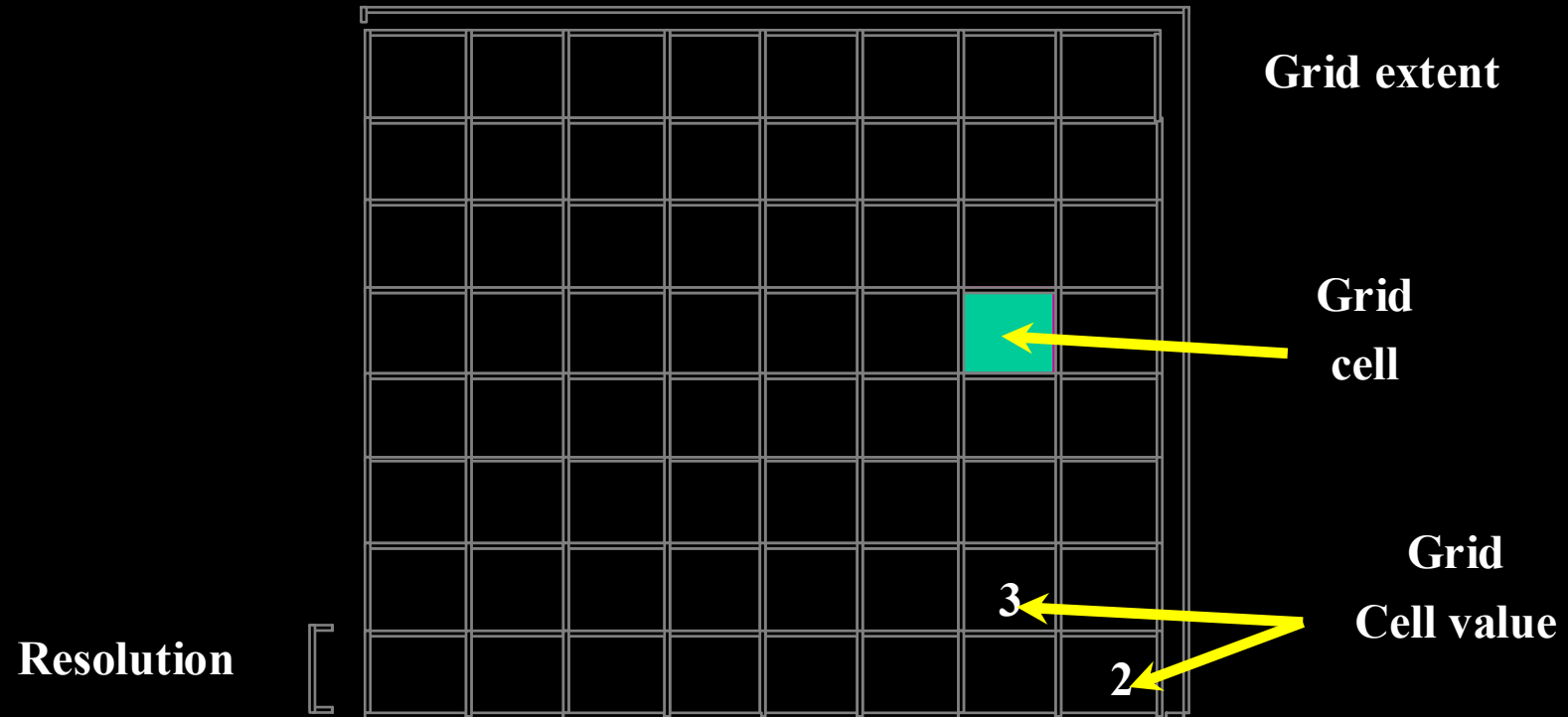
- modeling flow through the connecting lines in a network,
- combining adjacent polygons with similar characteristics,
- identifying adjacent features, and
- overlaying geographic features.

Raster Data Model

Raster Data Model

- Represents geographic data as images in a regular grid of cells.
- Row and column numbers define the locations of each cell in the grid or map. These cells are known as **pixels**. Each pixel can be assigned attribute values.
- Examples of typical raster data:
 - Remote sensing images
 - Scanned maps
 - Grids (ArcGIS & ArcInfo specific)
 - Graphical images (TIFF, JPEG, BMP, GIF, etc.)
 - SRTM DEM (Digital Elevation Model)

RASTER DATA MODELS

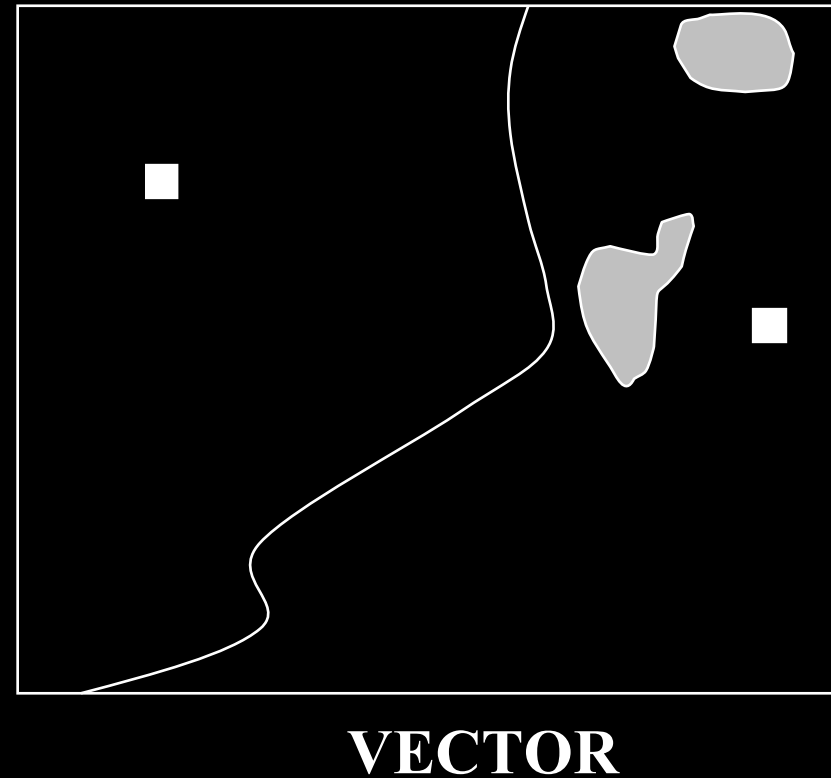
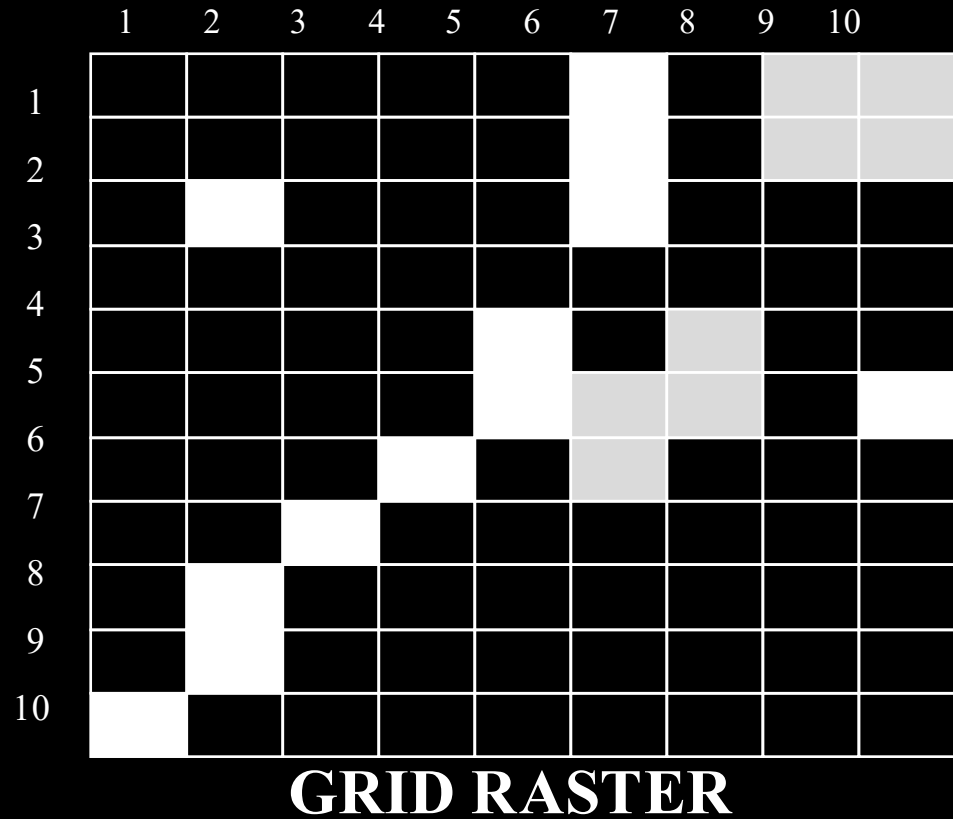


Generic structure for a grid.

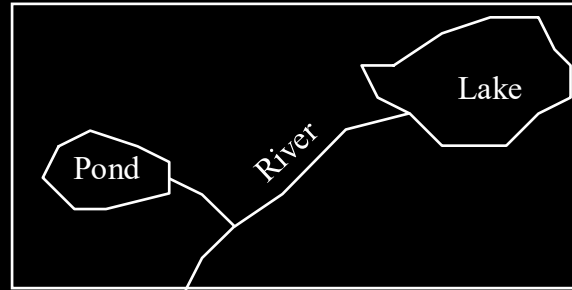
Raster Grid Properties

- Each Grid Cell holds one value even if it is empty.
- A cell can hold an index standing for an attribute.
- Cell resolution is given as its size on the ground.
- Point and lines move to the center of the cell.
- Minimum line width is one cell.
- Rasters are easy to read and write, and easy to draw/visualize on the screen.

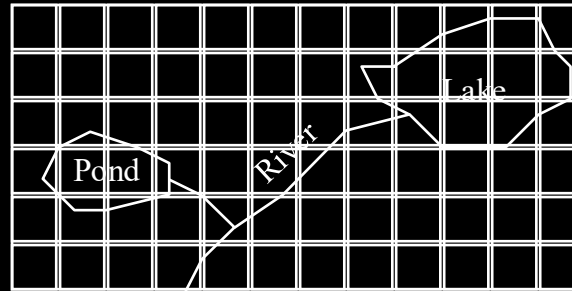
DATA MODEL OF RASTER AND VECTOR



Creating a Raster



Reality - Hydrography



Reality overlaid with a grid

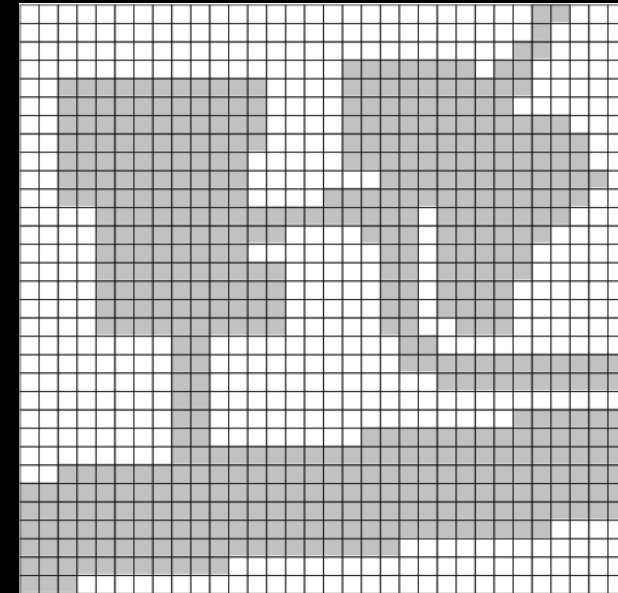
0	0	0	0	0	0	0	0	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1	1
0	0	0	0	0	0	0	2	1	1	1	0
0	1	1	2	0	0	2	0	0	0	0	0
0	0	0	0	2	2	0	0	0	0	0	0
0	0	0	2	0	0	0	0	0	0	0	0

Resulting raster

0 = No Water Feature
1 = Water Body
2 = River

Basic Aspects of Raster Structures

- Rasters provide a fixed grid for storing data
- Cells are addressed using the row and column number
- Rasters may be used to represent a range of computable spatial objects, including:
 - A point represented by a single cell
 - A strand or polyline represented by a sequence of neighboring cells
 - A connected area represented by a continuous collection of cells
- Rasters may be stored as arrays, which are natural computable structures, but can be wasteful in terms of space



Comparison of Raster & Vector Data Models

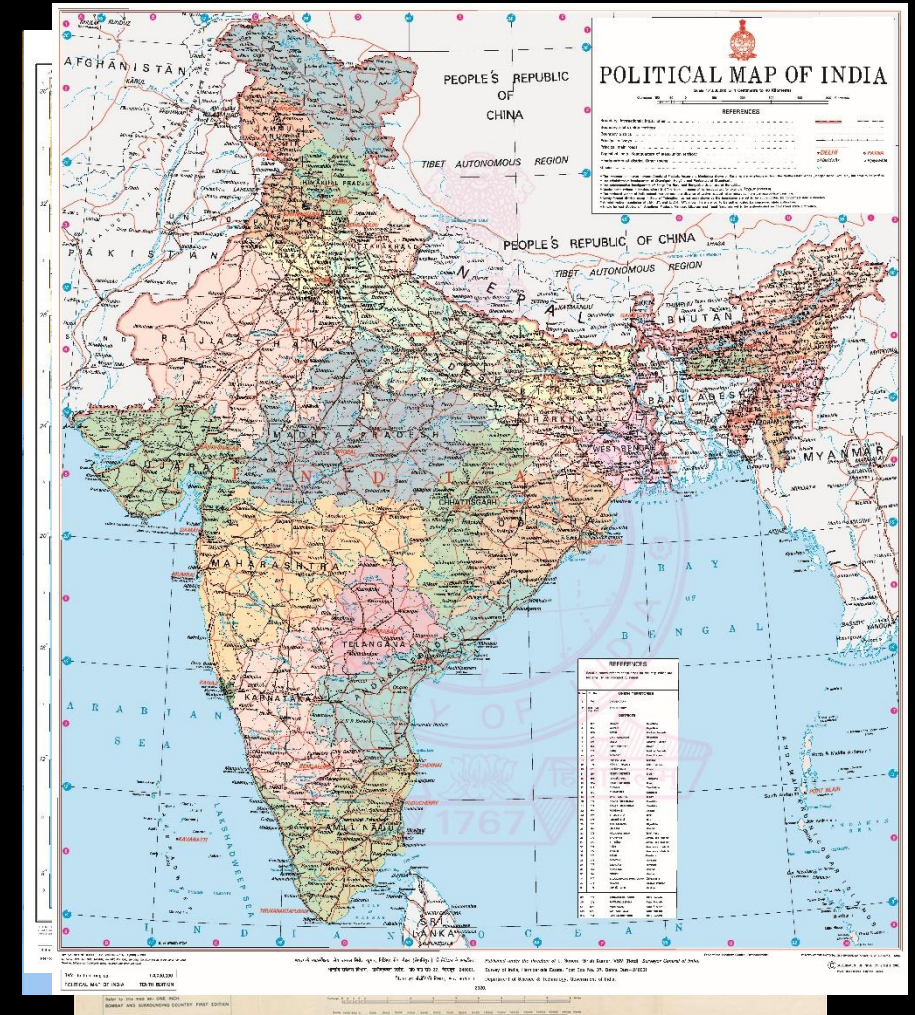
S. No.	Aspect	Raster	Vector
1	Framework	Cell	Point/Line/Polygon
2	Data structure	Simple	Complex
3	Data volume	Large	Small
4	Representation of real world features	Poorer	Better
5	Geometrical accuracy	Lower	Higher
6	Resolution	Limited by pixel size	Unlimited (in theory)
7	Recording of object attributes	Only one attribute per raster layer	Multiple attributes in one layer
8	Layers required	More	Few
9	Location and Attribute storage	Together	As separate tables
10	Topological Information	Not stored	Stored
11	Spatial modeling & overlay operations	Simpler	Harder
12	Network Analysis capabilities	Poor	Good
13	Processing speed	Fast	Slow
14	Conversion of remote sensing data	Often little or no conversion required	More processing required for conversion
15	Data Costs	More affordable	Less affordable

Part-1

Map- Basics **Shapes of the Earth, Datum**

What is a Map?

- It is the primary language of geography. Therefore, it is the fundamental language of digital geography.
- It is a **graphic form of spatial data abstraction**, composed of:
 - different grid systems,
 - projections,
 - symbol libraries,
 - methods of simplification & generalization and scales.
- Some of the common maps are:
 - Political Maps
 - Physical Maps
 - Climate Maps
 - Topographic Maps



Map Making Process

During the process of map making the cartographer must (based on Robinson et al., 1995):

- Establish the *purpose* the map is to serve.
- Define the *scale* at which the map is to be produced.
- Select the *features* from the real world which must be portrayed on the map.
- Choose a method for the *representation* of these features.
- *Generalize* these features for representation in 2D
- Adopt a *map projection* for placing these features onto a flat piece of paper.
- Apply a *spatial referencing* system to locate these features relative to each other
- *Annotate* the map with keys, legends and text to facilitate use of the map.

What is Map Generalization?

Generalization: All spatial data are generalization or implication of real world features. Generalization is needed because data are required at a particular scale.

Cartographic generalisation is a **simplification process of a spatial data** set (depending on: the purpose of a map and scale of the map)

Map generalization refers to resolving conflicts that arise on a map due to limited space, too many details, or too much information.

Generalization can take several forms:

- The simplification or symbolization of features for emphasis
- The masking or displacement of detail to increase clarity or legibility
- The selection of detail for inclusion or omission from the map
- The exaggeration of features for emphasis

Georeferencing of GIS Data

[Assigning Map Projection and Spatial Referencing]

- **Geographic location** is the element that distinguishes geographic information from all other types.
- So, methods for specifying location on the Earth's surface are essential to the creation of useful geographic information.
- Humanity has developed many such techniques over the centuries.
- The representation of the location of real-world features within the **spatial framework** of a particular coordinate system is called *Georeferencing*.

Some commonly used systems for Georeferencing

System	Domain of uniqueness	Metric?	Example	Spatial resolution
Placename	varies	no	London, Ontario, Canada	varies by feature type
Postal address	global	no, but ordered along streets in most countries	909 West Campus Lane, Goleta, California, USA	size of one mailbox
Postal code	country	no	93117 (US ZP code); WC1E6BT(UK unit postcode)	area occupied by a defined number of mailboxes
Telephone calling area	country	no	805	varies
Cadastral system	local authority	no	Parcel 01452854, City of Springfield, Mass, USA	area occupied by a single parcel of land
Public Land Survey System	Western USA only, unique to Prime Meridian	yes	Sec 5, Township 4N, Range 8E	defined by level of subdivision
Latitude/longitude	global	yes	119 degrees 45 minutes West, 34 degrees 40 minutes North	infinitely fine
Universal Transverse Mercator	zones six degrees of longitude wide, and N or S hemisphere	yes	563148E, 4356732N	infinitely fine
State Plane Coordinates	USA only, unique to state and to zone within state	yes	55088.34E, 75210.76N	infinitely fine

Source: Longley et al., (2005)

Elements of Spatial Framework

To represent the real-world locations within a spatial framework (e.g., a map), we must know the various elements of the framework.

Spatial framework includes:

- Earth Models
- Datum
- Coordinate Systems
- Map Projections

Earth Models

Modeling the shape of the Earth

The depiction of three-dimensional position requires a three dimensional surface.

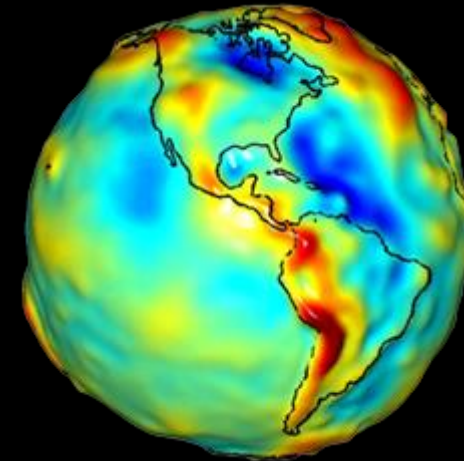
The First approximation and a convenient surface to represent the shape of the Earth is the geoid.

It is the *equipotential* surface of the earth's gravity field that on average coincides with mean sea level in the open oceans.

Due to variations in gravity, the geoid undulates significantly and a regular mathematical model is required for the calculations associated with a datum.

An appropriate mathematical model is an ellipsoid (or spheroid).

A depiction of the Western Hemisphere geoid.



Source: oceanservice.noaa.gov

What is an Equipotential Surface?

It is a surface having a constant potential and being everywhere perpendicular to the direction of gravity. The equipotential surface is level, i.e., the water is at rest.

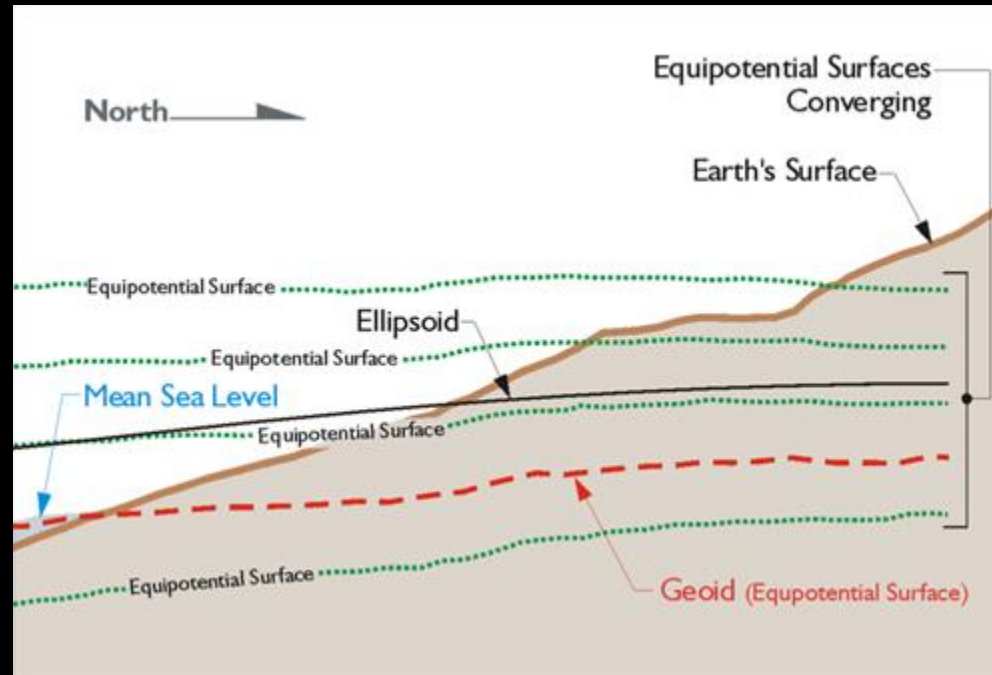
There exists an infinity of equipotential surfaces. These surfaces do not intersect each other, but they converge towards the poles. Thus, the geometric distance between two equipotential surfaces is less at the poles than at the equator.

The gravitational potential (V) is the potential energy (PE) per unit mass (m):

$$V = PE/m \text{ (Unit: m}^2/\text{s}^2\text{)}$$

Potential energy is equal (in magnitude, but negative) to the work done by the gravitational field moving a body to its given position in space from infinity.

Ellipsoid-Geoid-Mean Sea Level with Several Equipotential Surfaces



Practical Use of Equipotential Surfaces

Definition of the vertical (Local plumb line) → direction of the gravity field → perpendicular to equipotential surfaces.

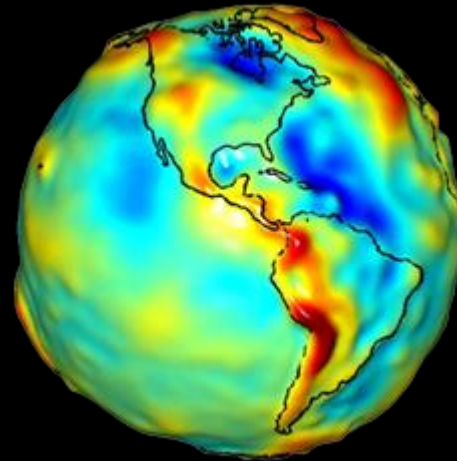
It is the zero surface to measure elevations.

Equipotential surfaces → define the horizontal reference for vertical measurements

The Geoid

The Geoid is a particular equipotential surface of the Earth gravity field that most **closely approximates the mean sea surface**.

The geoid is one such surface with a particular potential value.



The Western Hemisphere geoid.

Source: oceanservice.noaa.gov

For example, Natural Resource Canada (NRC) uses the equipotential surface ($62,636,856.0 \text{ m}^2/\text{s}^2$) representing best the coastal mean water level for North America as the new vertical reference system for Canada.

Geopotential value(s) of the Global Geoid

Every equipotential surface is associated with a single value of potential energy. The geodesy community denotes the potential value of the geoid as W_0

For example, (Amin, et al., 2019) defined the geoid to be $W_0 = 62,636,848.102 \pm 0.004 \text{ m}^2\text{s}^{-2}$ based on gravity observations obtained from satellites.

Another study by Dayoub et al (2011) geoid geopotential $W_0 = 62636854.29 \pm 0.02 \text{ m}^2\text{s}^{-2}$ based on EGM 2008 and Estimating the Circulation and Climate of the Ocean, phase II (ECCO-2) oceanographic model

Source:

Amin, H., et al (2019). A global vertical datum defined by the conventional geoid potential and the Earth ellipsoid parameters. *Journal of Geodesy*, Volume 93, pp. 1943-1961.

Dayoub et al (2011). The Geoid Geopotential Value for Unification of Vertical Datums. *FIG Working Week 2011*

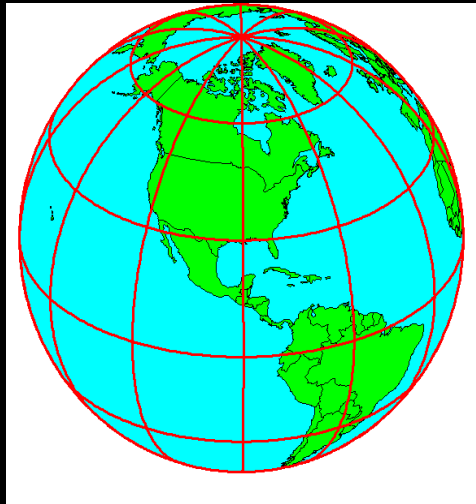
How can we make measurements on such a irregular figure?

- Surface approximated using mathematically smooth shapes.
- There are many models available such as
 - Flat earth model
 - Spherical model- Eratosthenes
 - Ellipsoidal model- Sir Isaac Newton

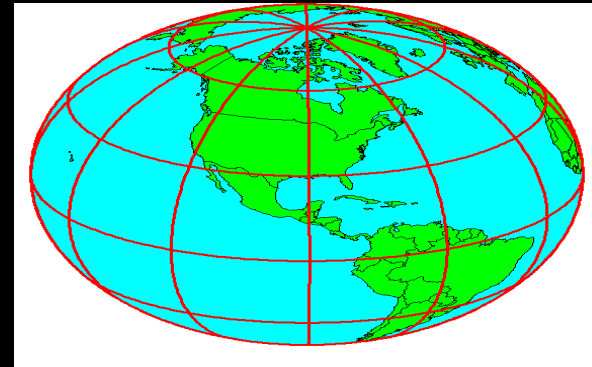
Shape of the Earth

We think of the
earth as a **sphere** ...

Ex: Globe



when it is actually an **oblate spheroid**
(ellipsoid), slightly larger in radius at
the equator than at the poles.



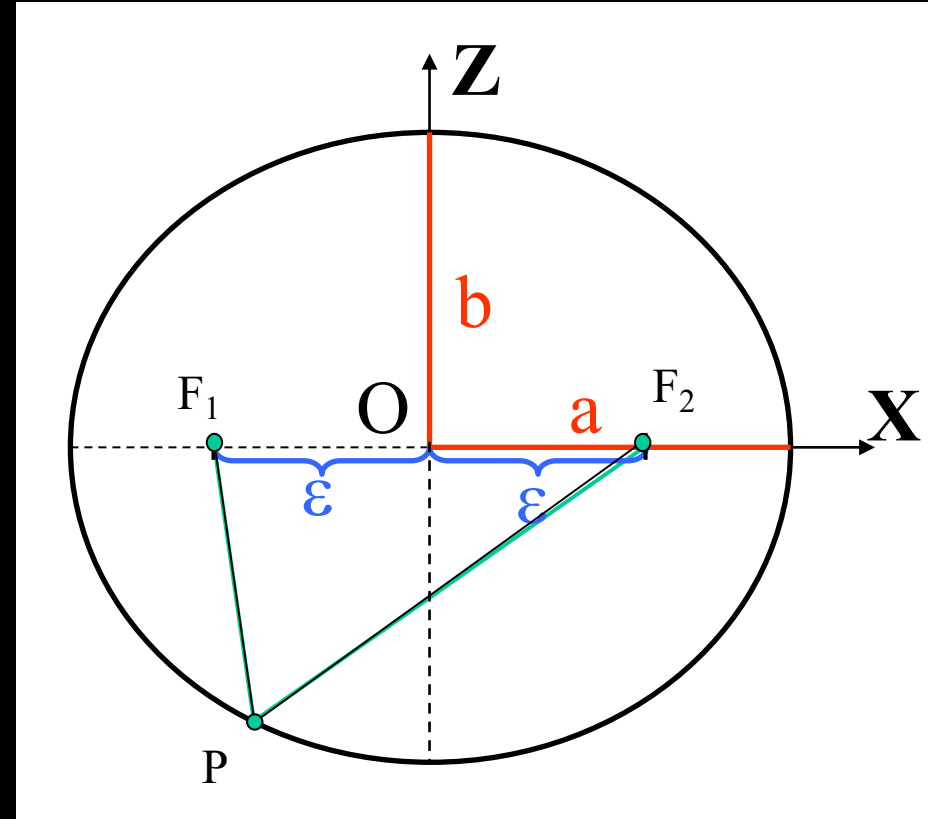
Ellipse

An ellipse is defined by:

- Focal length = ε
- Flattening ratio: $f = (a-b)/a$
- Distance F_1 -P- F_2 is constant for all points P on ellipse
- When $\varepsilon = 0$, ellipse = circle

For the Earth:

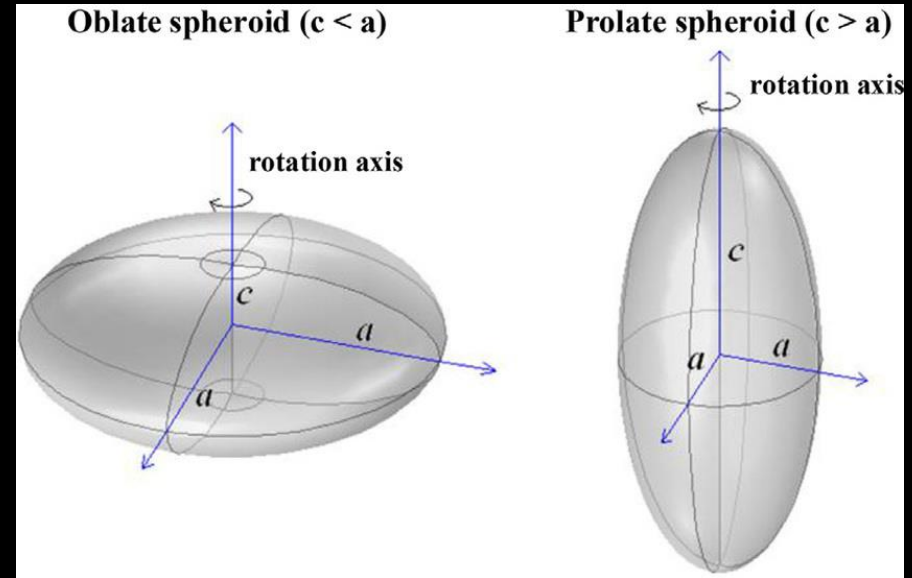
- Major axis: $a = 6378$ km
- Minor axis: $b = 6357$ km
- Flattening ratio: $f = 1/300$



Ellipsoid

Rotate an ellipse around one of its axes.

- Semi-minor axis: Prolate Spheroid
- Semi-major axis: Oblate Spheroid



Oblate Spheroid (aka Ellipsoid) closely matches the actual shape of the Earth.

Ellipsoid Models of the Earth

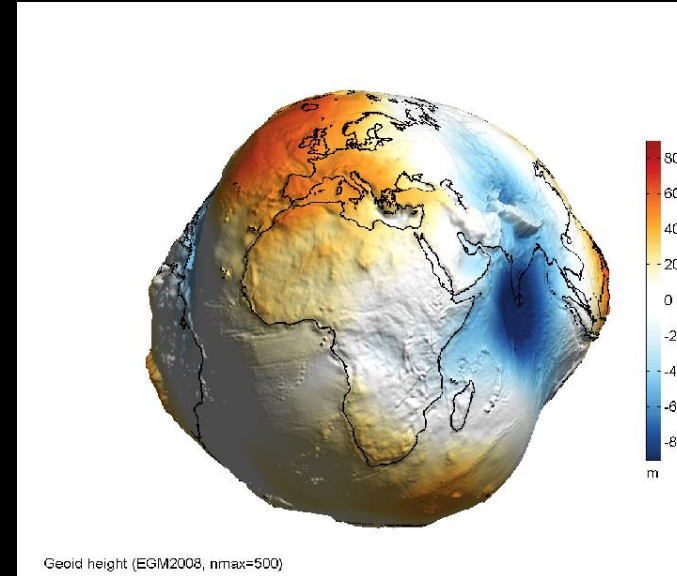
- Ellipsoidal models aid geographical analysis by providing a **smooth and simplistic mathematical representation** of the Earth surface.
- This helps us to measure distances on the surface without having to account for terrain elevation changes.
- Based on the mathematical properties adopted, several reference ellipsoids are in use.

Reference ellipsoid name	Equatorial radius (m)	Polar radius (m)	Inverse flattening	Where used
Maupertuis (1738)	6,397,300	6,363,806.283	191	France
Plessis (1817)	6,376,523.0	6,355,862.9333	308.64	France
Everest (1830)	6,377,299.365	6,356,098.359	300.80172554	India
Everest 1830 Modified (1967)	6,377,304.063	6,356,103.0390	300.8017	West Malaysia & Singapore
Everest 1830 (1967 Definition)	6,377,298.556	6,356,097.550	300.8017	Brunei & East Malaysia
Airy (1830)	6,377,563.396	6,356,256.909	299.3249646	Britain
Bessel (1841)	6,377,397.155	6,356,078.963	299.1528128	Europe, Japan
Clarke (1866)	6,378,206.4	6,356,583.8	294.9786982	North America
Clarke (1878)	6,378,190	6,356,456	293.4659980	North America
Clarke (1880)	6,378,249.145	6,356,514.870	293.465	France, Africa
Helmert (1906)	6,378,200	6,356,818.17	298.3	Egypt
Hayford (1910)	6,378,388	6,356,911.946	297	USA
International (1924)	6,378,388	6,356,911.946	297	Europe
Krassovsky (1940)	6,378,245	6,356,863.019	298.3	USSR, Russia, Romania
WGS66 (1966)	6,378,145	6,356,759.769	298.25	USA/DoD
Australian National (1966)	6,378,160	6,356,774.719	298.25	Australia
New International (1967)	6,378,157.5	6,356,772.2	298.24961539	
GRS-67 (1967)	6,378,160	6,356,774.516	298.247167427	
South American (1969)	6,378,160	6,356,774.719	298.25	South America
WGS-72 (1972)	6,378,135	6,356,750.52	298.26	USA/DoD
GRS-80 (1979)	6,378,137	6,356,752.3141	298.257222101	Global ITRS ^[3]
WGS-84 (1984)	6,378,137	6,356,752.3142	298.257223563	Global GPS
IERS (1989)	6,378,136	6,356,751.302	298.257	
IERS (2003) ^[4]	6,378,136.6	6,356,751.9	298.25642	^[3]

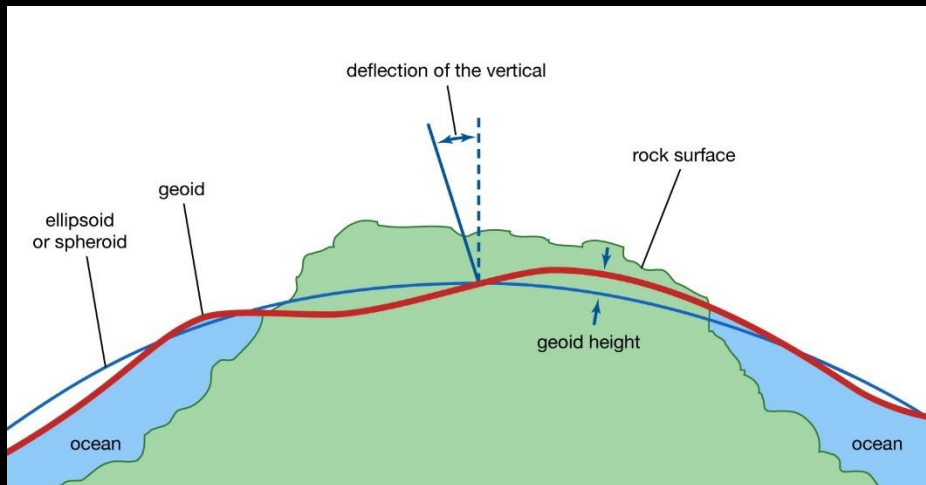
Comparison between Geoid and reference Ellipsoids

Global map with geoid heights of the *EGM 2008 gravity field* model, computed relative to the *GRS80 ellipsoid*.

The elevation difference between geoid surface and reference ellipsoid surface is called **Geoid Height**.

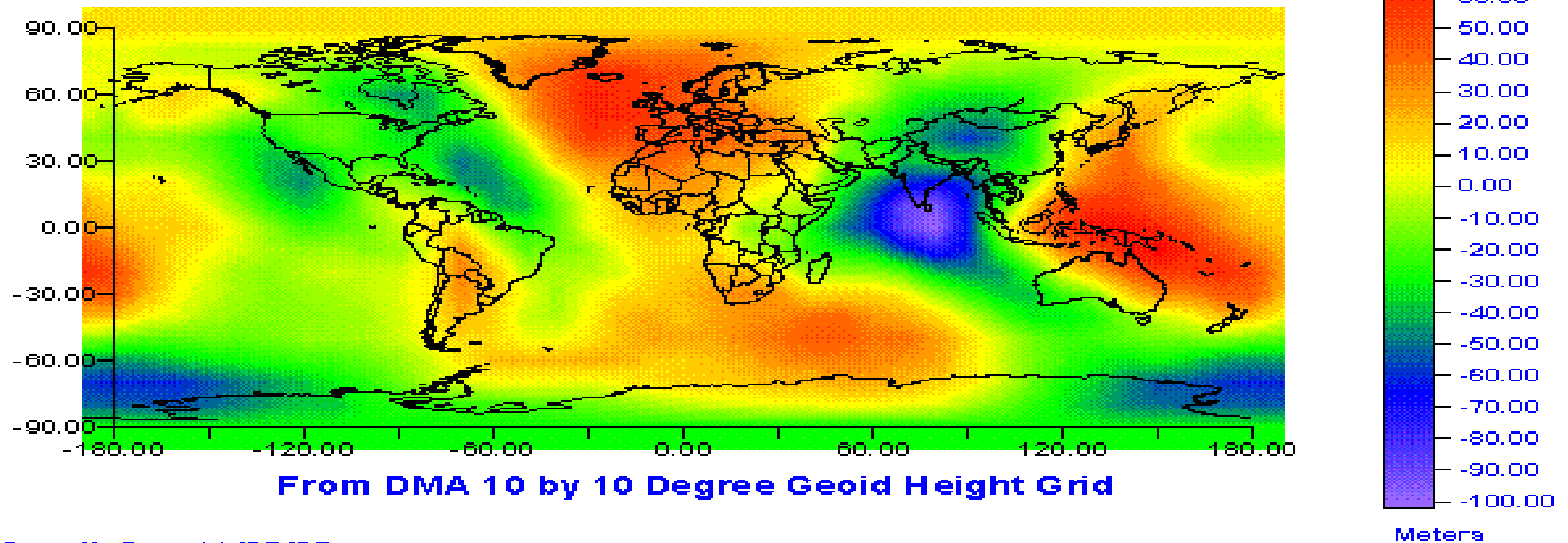


Source: GFZ Potsdam



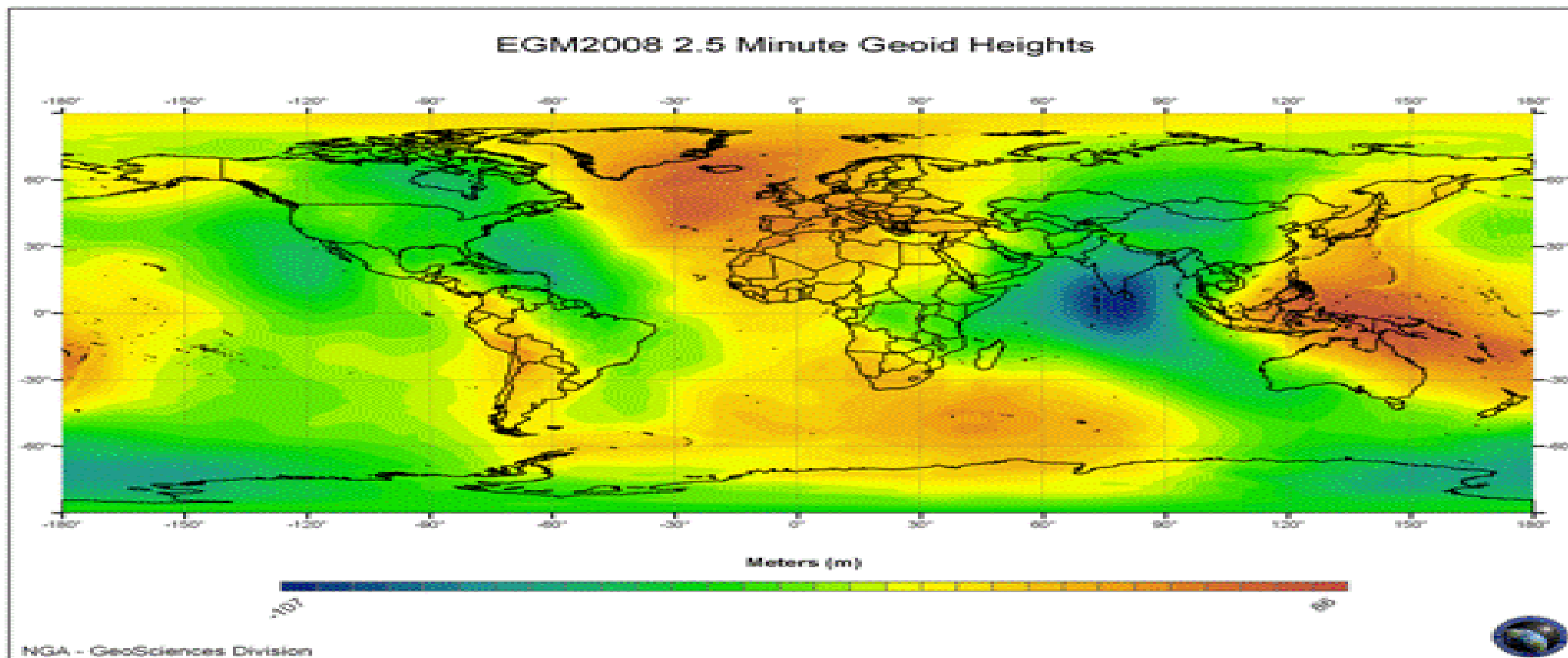
Earth Gravitational Model (EGM) 1996 Geoid heights based on WGS 84 reference spheroid

WGS-84 Geoid Height



Peter H. Dana 11/05/95

EGM 2008 Geoid heights based on WGS 84 Ellipsoid



Datum

What is a Datum?

Generally, **Datum** refers any numerical or geometrical quantity or set of such quantities which serve as a reference or base for other quantities.

Geodetic Datum

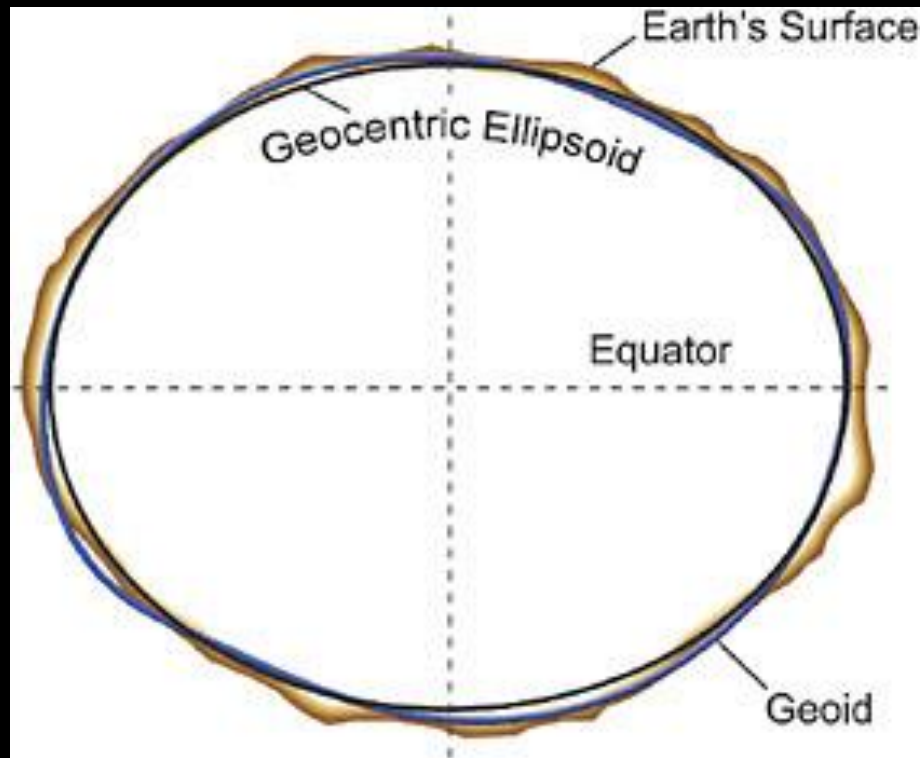
- A geodetic datum defines the size and shape of the earth, and the origin and orientation of the axis used to define the location of points.
- Datum serves as a *reference frame for measurements (both horizontal and vertical)* on the Earth surface in which the coordinates of spatial data may be expressed.
- Datum comprises both horizontal and vertical reference frames.

Horizontal datum : Any reference ellipsoids

Vertical datum: Geoid , Mean Sea Level (MSL)

Horizontal Datum

A reference ellipsoid and its orientation used to define a horizontal coordinate system is called as **Horizontal Datum**.



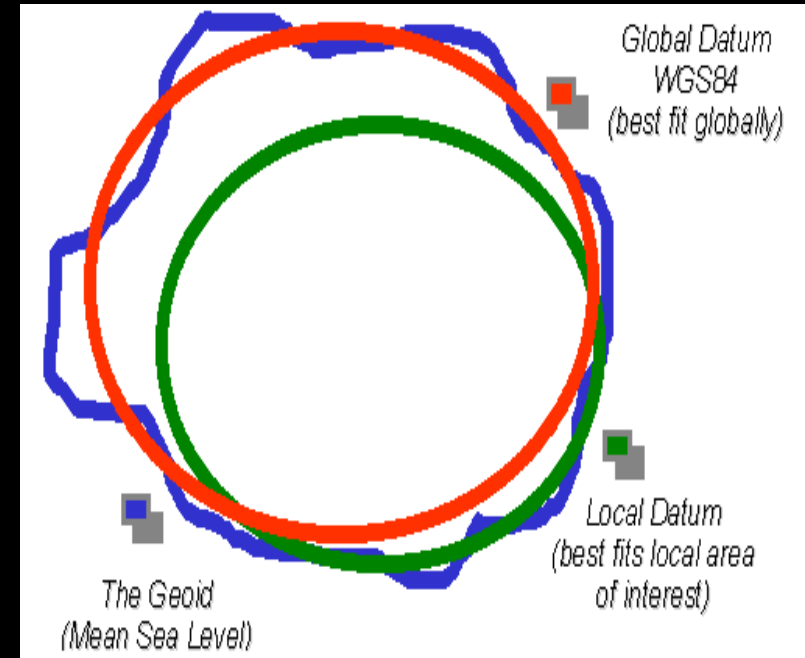
Types of Horizontal Datum

Geocentric Datum

- Geocentric (earth-centered) datum uses the earth's center of mass as the origin.
- The most recent and widely used geocentric datum is **WGS 1984**.
- It serves as the framework for locational measurement worldwide.

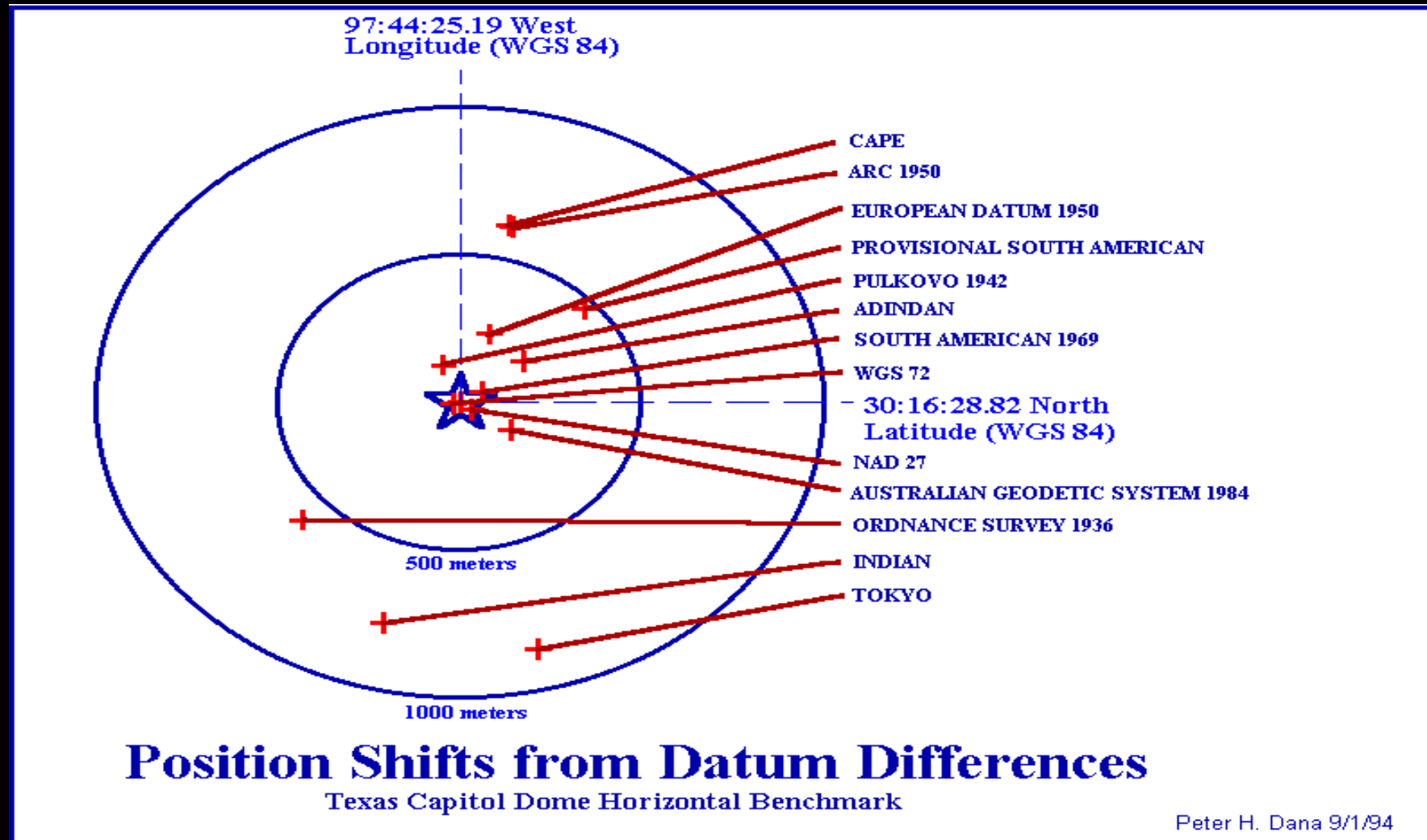
Local Datum

- A local datum aligns its spheroid to closely fit the earth's surface *in a particular area*.
- A point on the surface of the spheroid is matched to a particular position on the surface of the earth. This point is known as the origin point of the datum.
- The coordinates of the origin point are fixed, and all other points are calculated from it.
- Example: **Everest 1956 datum for India**



Source: International Federation of Surveyors

Position Accuracy due to Datum Shift



Horizontal Datum used in India (till early 2000s)

Datum Name: Indian Geodetic Datum

Reference Ellipsoid: Everest 1956

Semi-major axis- 6377301.243 m

Semi-minor axis- 6356100.228 m

Flattening ratio- $1/300.8017$

Origin: Kalyanpur Madhya Pradesh, India

[Geoidal height at this location was assumed to be zero]

Horizontal Datum used in India (after 2005)

Datum Name: WGS 1984 Datum

Reference Ellipsoid: WGS 1984 Ellipsoid

Semi-major axis- 6378137.0 m

Semi-minor axis- 6356752.314140 m

Flattening ratio- $1/298.25722$

Origin: Center of the Mass of the Earth

Note:

The geoid generally differs from an earth-centered horizontal geodetic datum by <100 m.
Currently, the Survey of India (SoI) uses WGS 84 datum for all their open series maps.

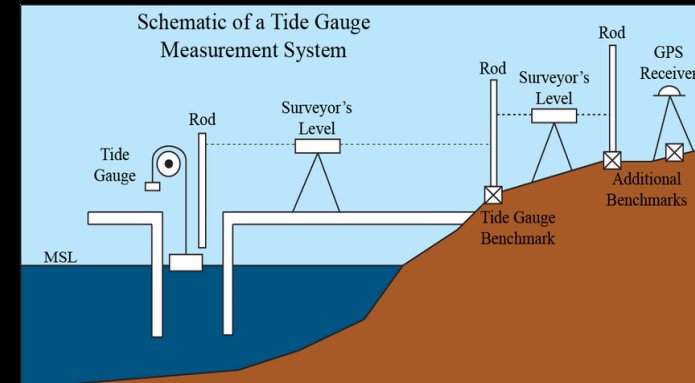
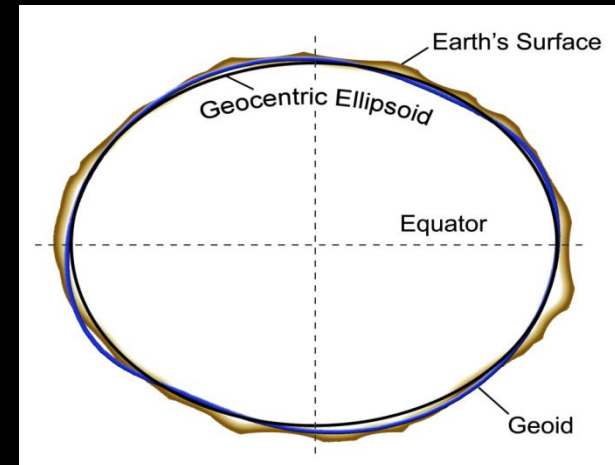
Vertical Datum

A vertical datum defines the origin for height or depth values.

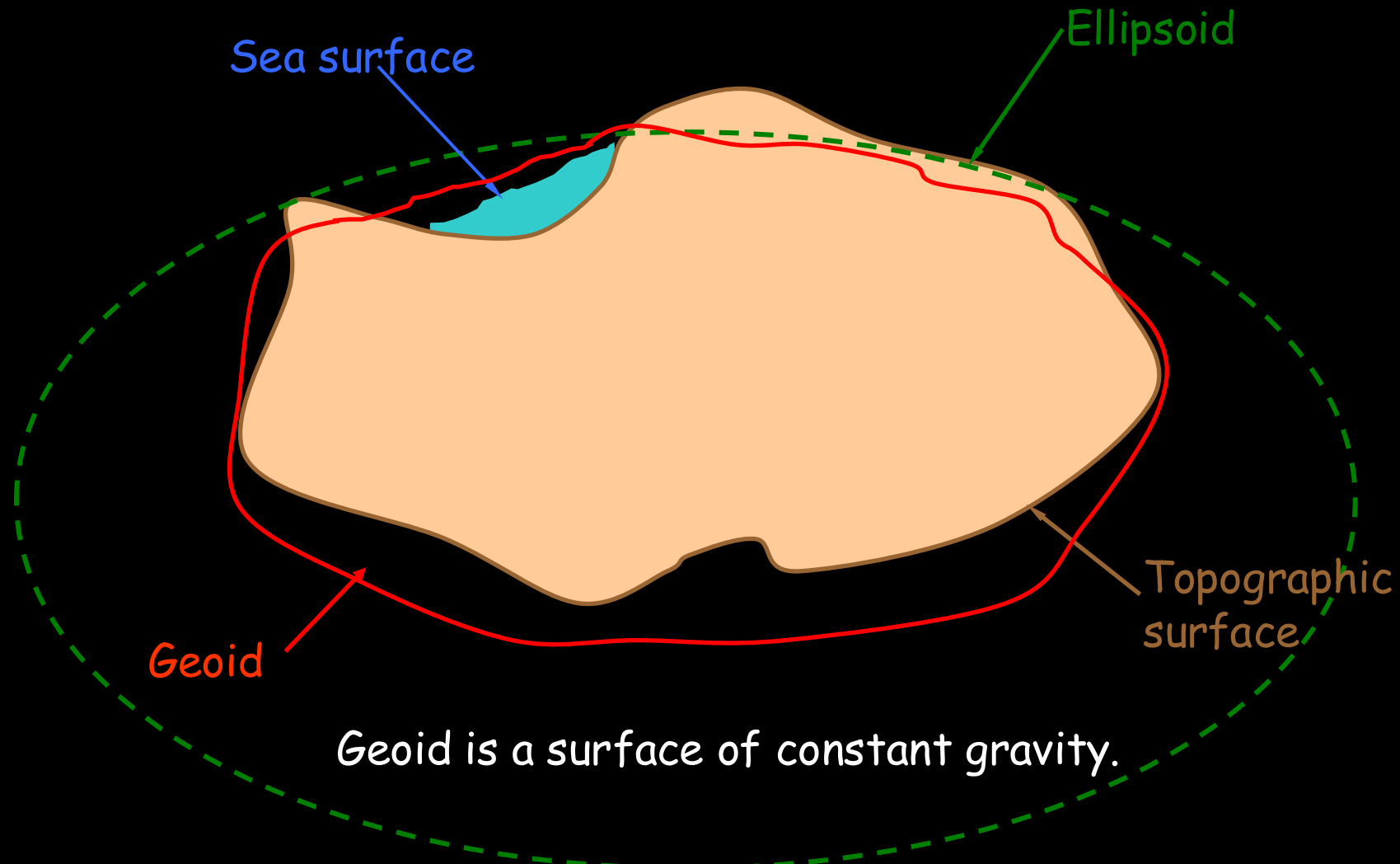
Three different types of datum surfaces:
Spheroidal (ellipsoidal) Ex: WGS 84

Gravity-related (geoidal) Ex: Orthometric heights

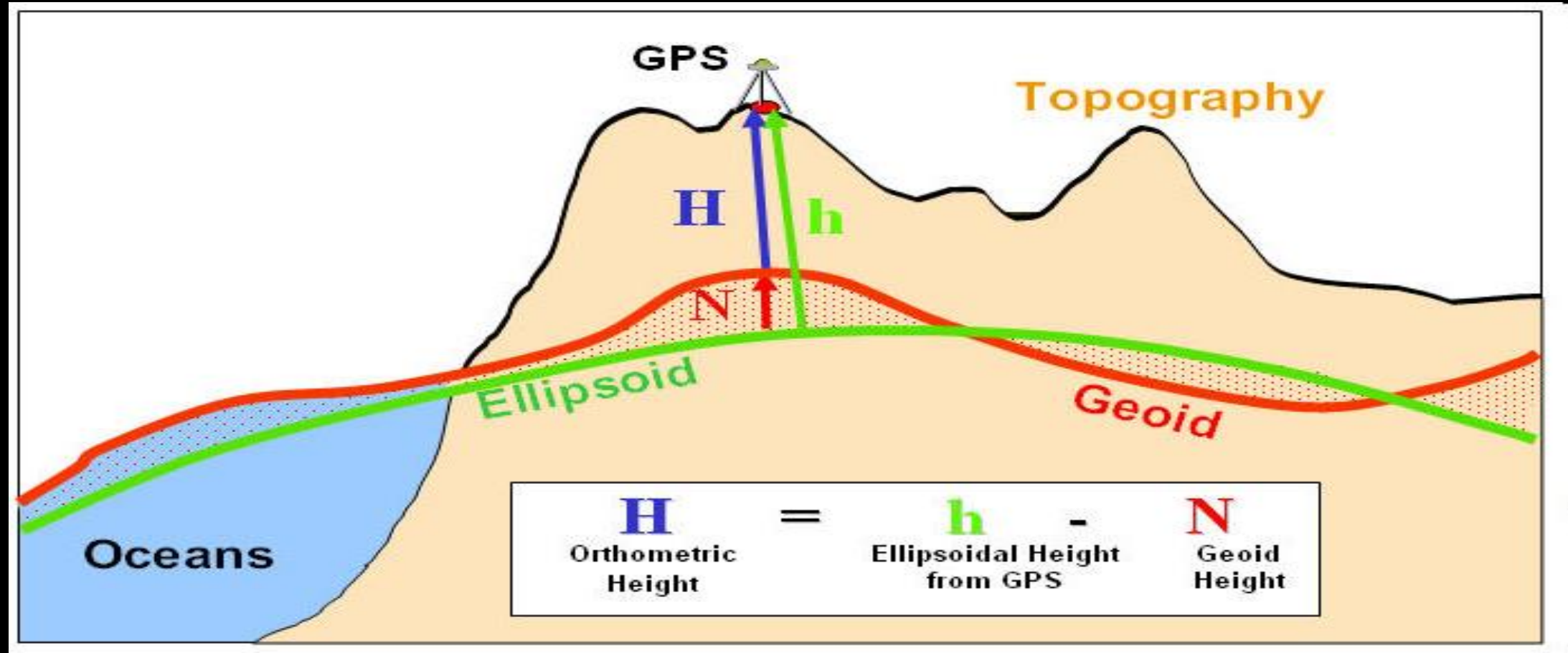
Tidal Heights Ex: Mean Sea Level (MSL)



Different representation of Earth Surfaces



Elevation Measurements



Commonly used Datums

- WGS 84, 72, 66 and 60 of the World Geodetic System
- NAD83, the North American Datum which is very similar to WGS 84
- OSGB36 of the Ordnance Survey of Great Britain
- ETRS89, the European Datum, related to ITRS
- GDA94, the Australian Datum
- JGD2011, the Japanese Datum
- BJS54 and XAS80, old geodetic datum used in China
- GCS-02 and BD-09, Chinese encrypted geodetic datum.
- PZ-90.11, the current geodetic reference used by GLONASS
- GTRF, the geodetic reference used by Galileo; currently defined as ITRF2005
- CGCS2000, or CGS-2000, the geodetic reference used by BeiDou Navigation Satellite System; based on ITRF97
- International Terrestrial Reference Frames (ITRF88, 89, 90, 91, 92, 93, 94, 96, 97, 2000, 2005, 2008, 2014), different realizations of the ITRS.

About WGS 84

WGS 84 is an Earth-centered, Earth-fixed terrestrial reference system and geodetic datum.

WGS 84 is based on a consistent set of constants and model parameters that describe the Earth's size, shape, and gravity and geomagnetic fields.

WGS 84 is the standard U.S. Department of Defense definition of a global reference system for geospatial information and is the reference system for the Global Positioning System (GPS). It is compatible with the International Terrestrial Reference System (ITRS).

Part-2

Coordinate Systems

Coordinate Systems in Geography

- A grid (coordinate) system allows the **location of a point on a map or on the surface of the earth** to be described in a way that is meaningful and universally understood.
- Essential for: map making, and spatial analysis.



Common examples:

- Geographic Coordinate System (Latitude and Longitude)
- Universal Transverse Mercator (UTM) coordinate system

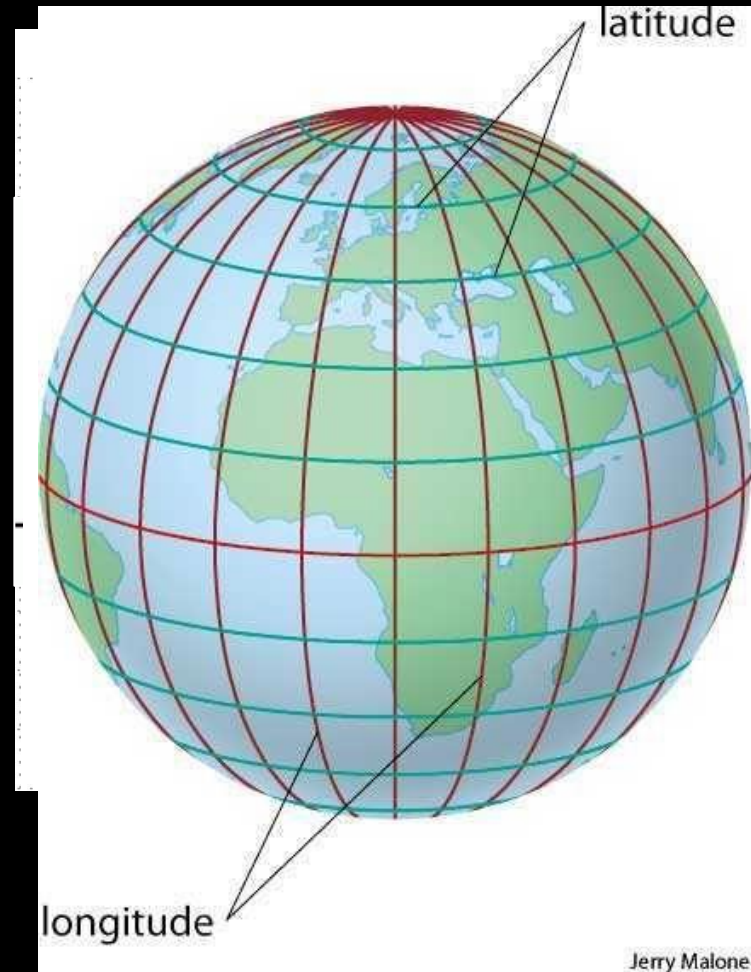
Basic Categories of Coordinates System in Geography

2-dimensional coordinate system:

- **Plane Cartesian** $\rightarrow (x, y)$
[aka Rectangular Coordinate System]
- **Plane Polar** $\rightarrow (r, \theta)$

3-dimensional coordinate system:

- **Cartesian** $\rightarrow (x, y, z)$
- **Geographic** $\rightarrow (\text{Latitude}, \text{Longitude})$



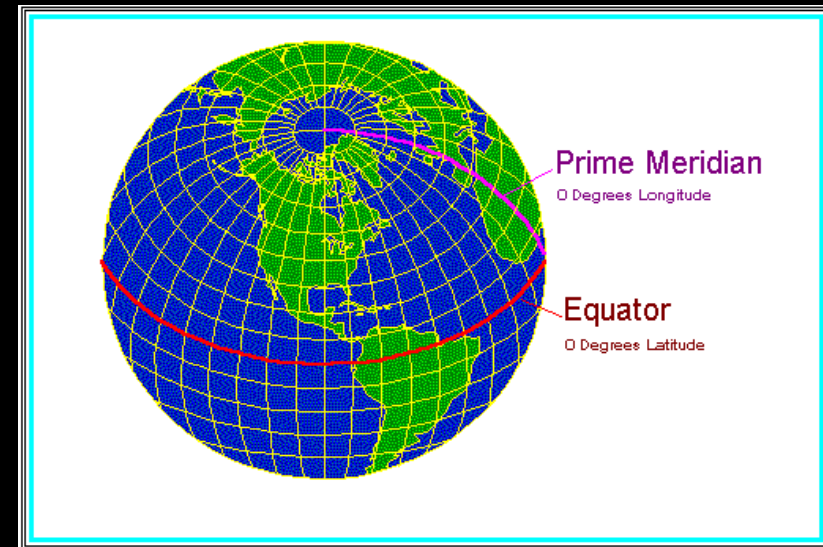
Geographic Coordinates

The geographic coordinates are often chosen such that one of the numbers represents vertical position, and two of the numbers represent horizontal position on a sphere or an ellipsoid. A common choice of coordinates is latitude, longitude and elevation.

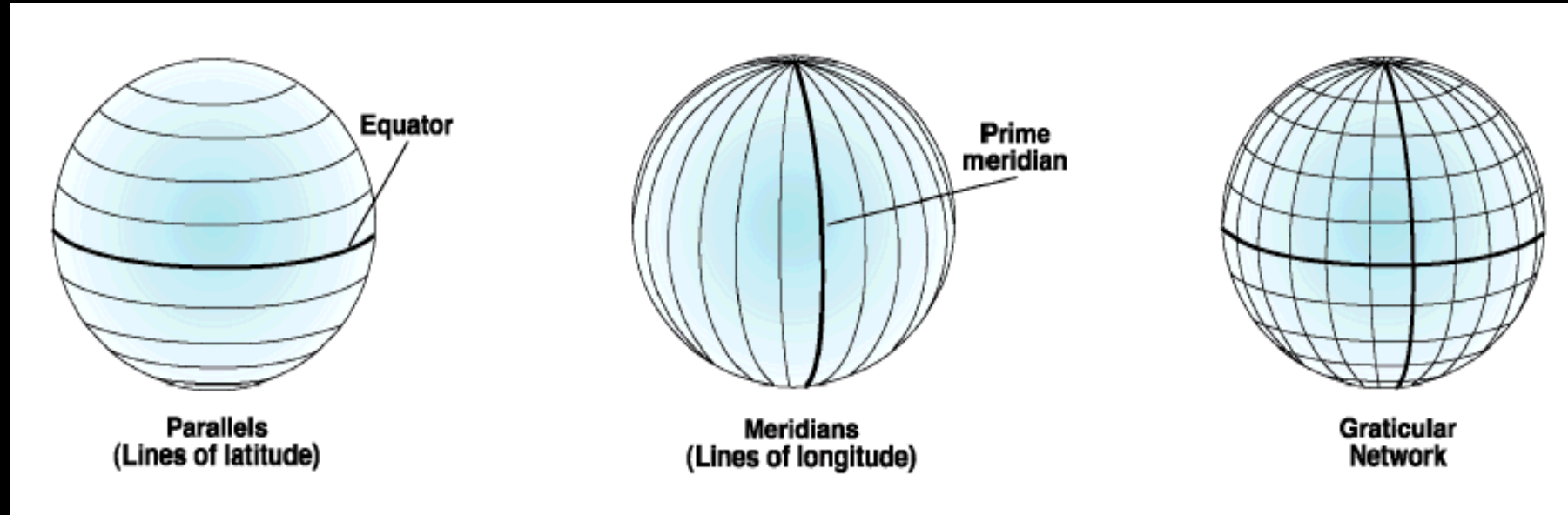
The lines from pole to pole are lines of constant **longitude**, or **meridians**.

The circles parallel to the equator are lines of constant **latitude**, or **parallels**.

The graticule determines the latitude and longitude of position on the surface.



Components of Geographic Coordinate System



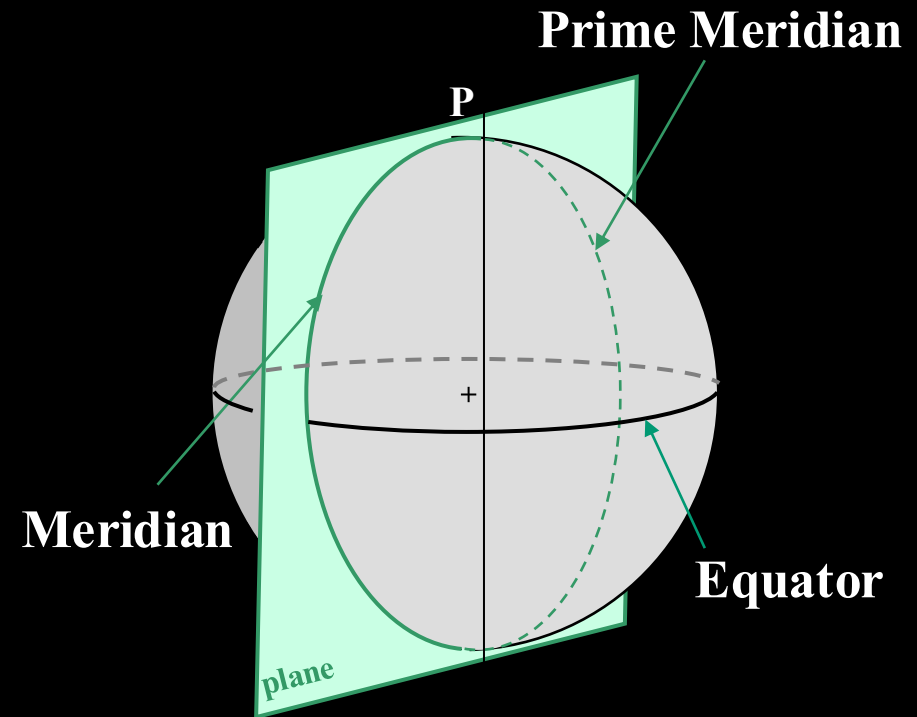
- Latitude and longitude are **angular measurements** made from the center of the earth to a point on the surface of the earth.

Latitudes/Parallels – east to west – varies from 0° to 90° – 0° at the Equator
Longitudes/Meridians – north to south – varies from 0° to 180° – 0° at the Prime Meridian

- Latitudes and Longitudes together constitute the **graticular network**.

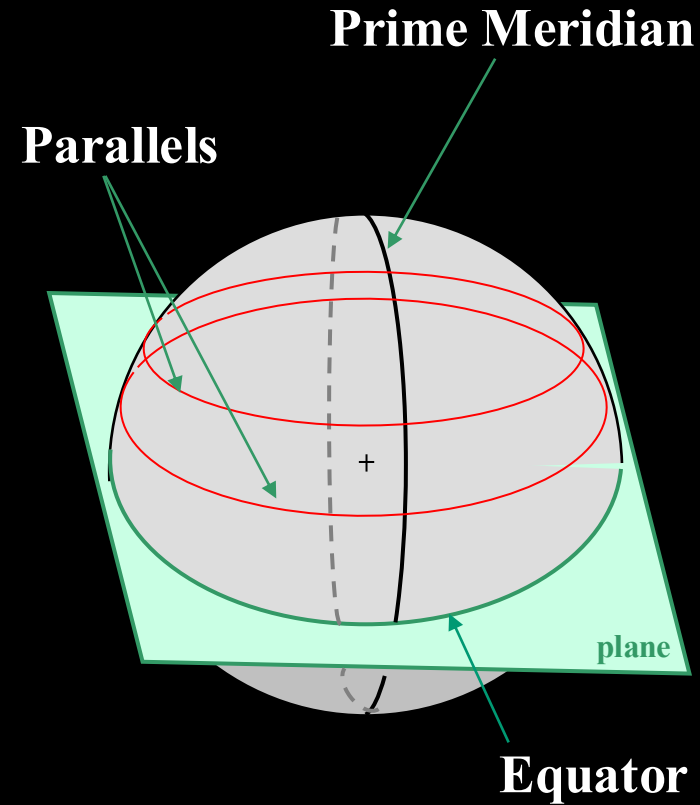
Meridians

- *Imagine a plane* that cuts through the center of the Earth.
- Intersection of this plane with the earth's surface → great circle
- Any half of a great circle that joins the poles → longitude/meridian
- The prime meridian (0°) runs through Greenwich, England.
- Longitude is recorded as degrees E/W of the prime meridian (up to 180°)



Parallels

- Are at right angles to the meridians
- The great circle (at the equator) is located halfway between the poles.
- Several small circles (called lines of latitude/parallels) are placed parallel to the equator.
- Latitude is recorded as degrees N/S of the equator (up to 90°)



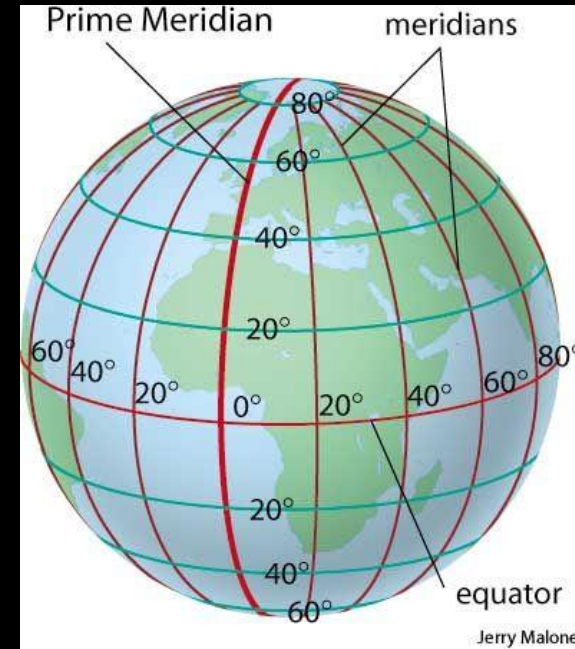
Graticules

- The pattern of meridians and parallels on the earth is a *graticule*.
- It has the following characteristics:
 - Parallels are equally spaced between the equator and the poles
 - Any two parallels are always the same distance apart
 - Meridians are spaced farthest apart at the equator and converge to a single point at the poles
 - Parallels and meridians cross one another at right angles

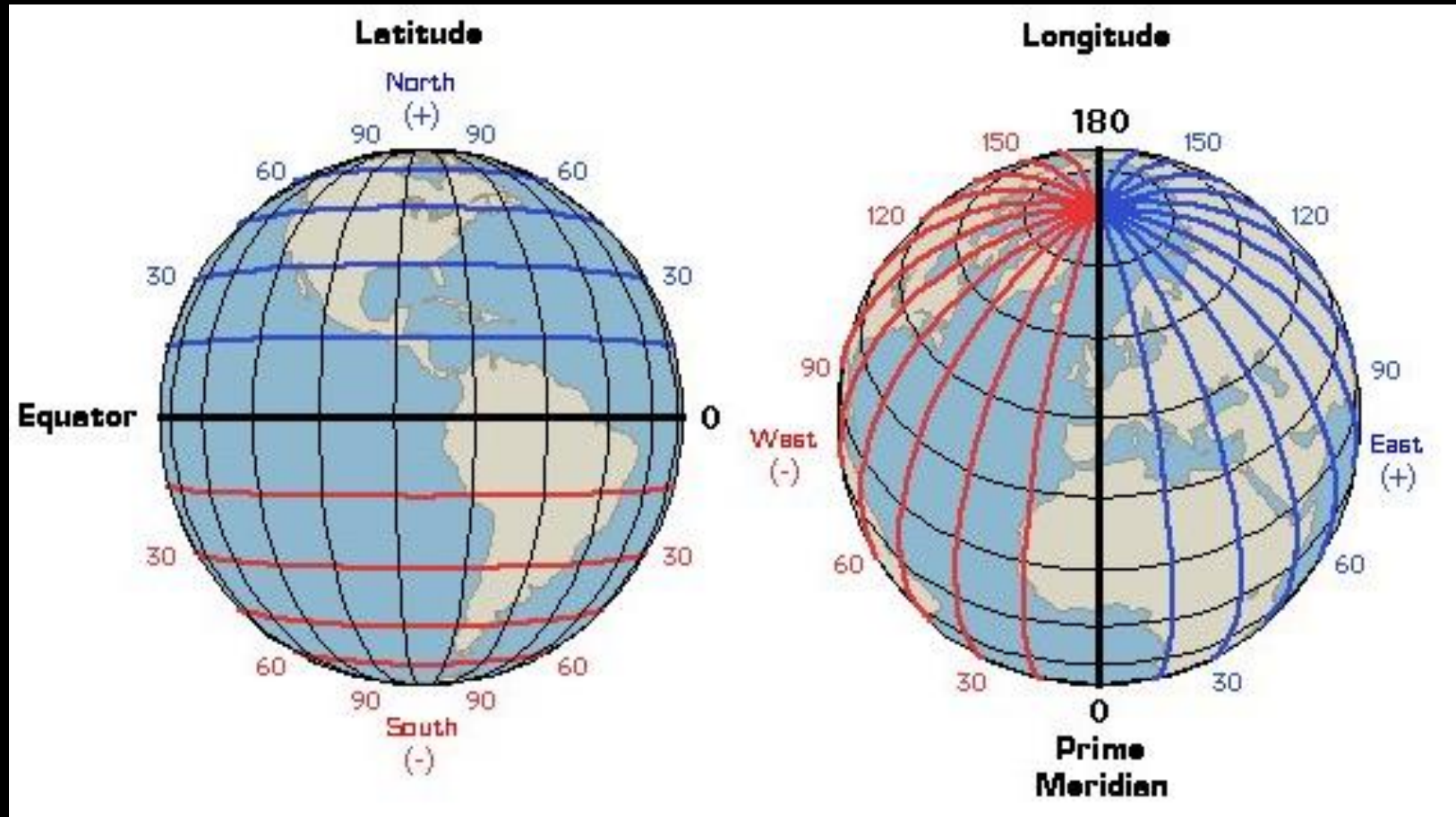


Significance of the Prime Meridian

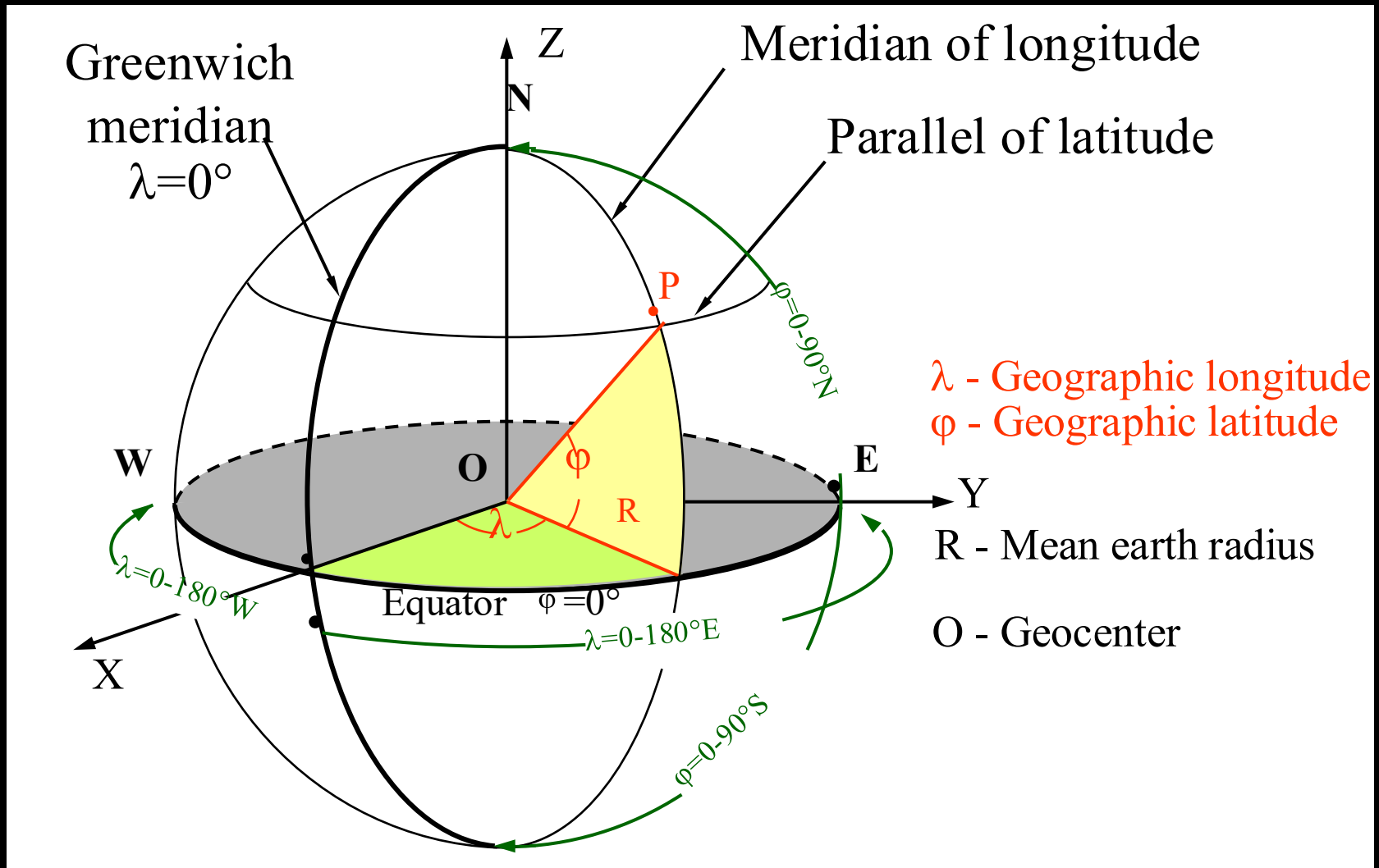
- It is an arbitrary line unlike the equator (which is determined by the axis of rotation).
- Various conventions have been used or advocated in different regions and throughout history.
- By International convention, the meridian passing through the *Royal Observatory, Greenwich, England*, was chosen as the Prime Meridian.
- At this meridian, the longitude is regarded as 0° .
- Together with its counterpart on the other side of the globe (180° meridian), it forms the great circle.
- This great circle divides the Earth, into the Eastern and Western hemispheres.



Pictorial representation of Equator, North & South Poles with Latitudes and Longitudes



Latitude and Longitude on a Sphere



Relation between Longitude and Time Zones

- The Sun completes its apparent revolution of 360° in 24 hours.
This means that the Sun crosses each of the 360 meridians of longitude once every 24 hours.
- In 1 hour, the Sun appears to move 15° ,
- In 4 minutes, it appears to move 1° arc,
- In 1 minute it appears to move $15'$ arc,
- In 4 seconds it appears to move $1'$ arc.
- From this, it becomes obvious that there is a direct relationship between arc and time such that 1 minute of time equals 15 minutes of arc.

Mean Solar Time is directly related to longitude. Your longitude = [(your Mean Solar Time) minus (Greenwich Mean Solar Time)] x 15 degrees.

Every 15° longitude is a time zone.

Units of latitude and Longitude

Latitude and Longitudes are measured in:

- a) degree, minutes and seconds format, or
- b) decimal degrees format, or

Degree, minutes and seconds format:

In this format, latitudes and Longitudes are expressed in **degrees° minutes' seconds''**

(Note: $1^\circ = 60'$ & $1' = 60''$)

Example: Location of Point X → $32^\circ 59' 16.0798\text{N}$ $96^\circ 44' 56.9522\text{W}$

Decimal degrees format:

Decimal degrees (dd) is the best suited for GIS.

$$\text{dd} = \text{d}^\circ + \text{m}'/60 + \text{s}''/3600$$

Carry enough decimal points for accuracy! → 6 decimals give 4 inch (10cm) accuracy

(Note: 1 meter= 3.2808 feet)

But one must use double precision storage → single precision accurate to only 2m.

Example: Location of Point X→ 32.98779994N 96.74915339W

(8 decimals-->1 millimeter accuracy!)

Length of one degree on the Earth Sphere

Latitude:

One degree of latitude measured along a meridian equals approximately 69 miles (111 km).

One minute is just over a mile (~1.85 km), and one second is around 100 feet (~30 m) (a pretty precise location on a globe with a circumference of 40,000 km).

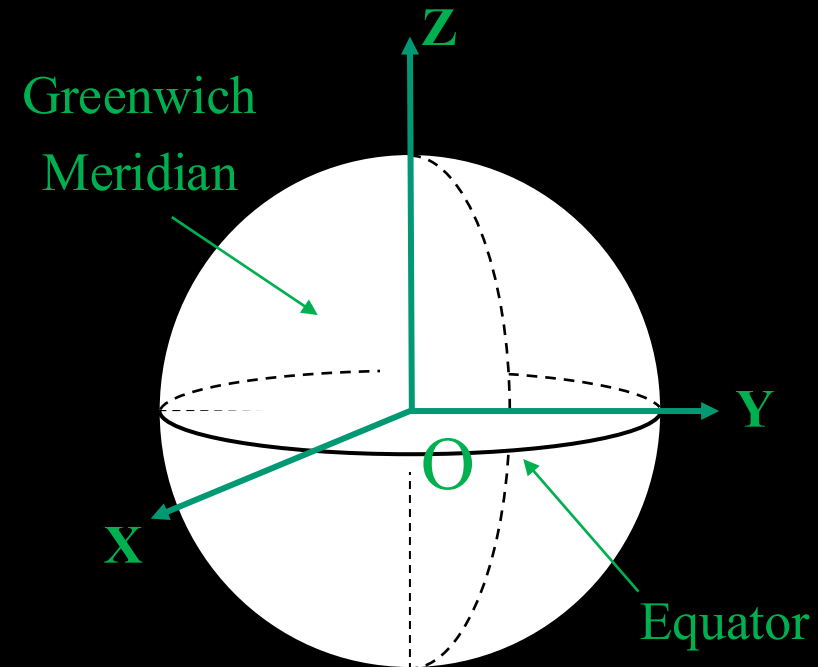
Longitude:

Because meridians converge at the poles, the length of a degree of longitude varies, from 69 miles at the equator to 0 at the poles (longitude becomes a point at the poles).



Global Cartesian Coordinates (x,y,z) System

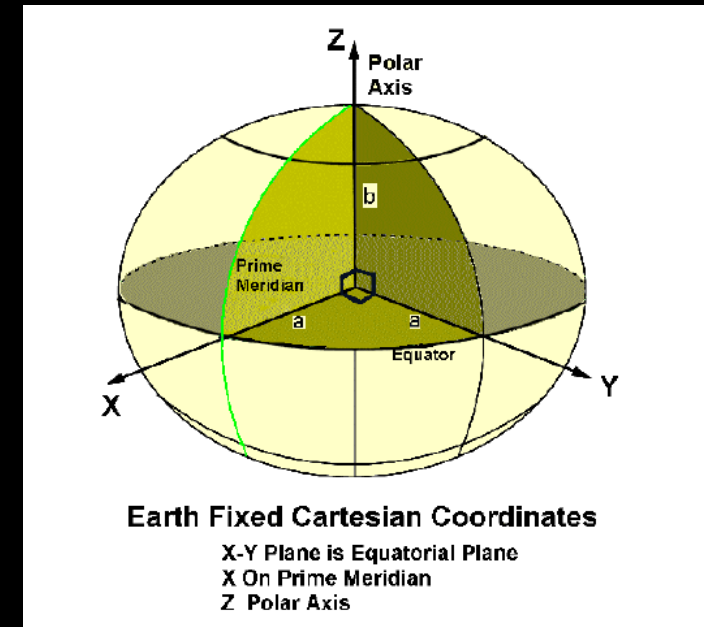
- It is a simple Cartesian coordinate system with the center of the earth at its origin.
- The +X axis passes through the Equator and Prime Meridian intersection.
- The +Z axis passes through the North Pole.
- The +Y axis is orthogonal to +X and +Z.
- As a result, this coordinate system rotates with the earth.
- The distances used along each axis are in meters.



Earth Centered, Earth Fixed (ECEF) Coordinate System

- These **Earth Centered, Earth Fixed ECEF** coordinates are the ones used by most satellites systems to designate an earth position.
- Only the center of the earth and the orientation of the axis is needed.
- Some high precision applications remain in ECEF to avoid additional error.

In fact, the Global Positioning System (**GPS**) uses **ECEF as it's primary coordinate system** and derives all other coordinates from it.



Part-3

Map Projections

Map Projections

Map projections refer to:

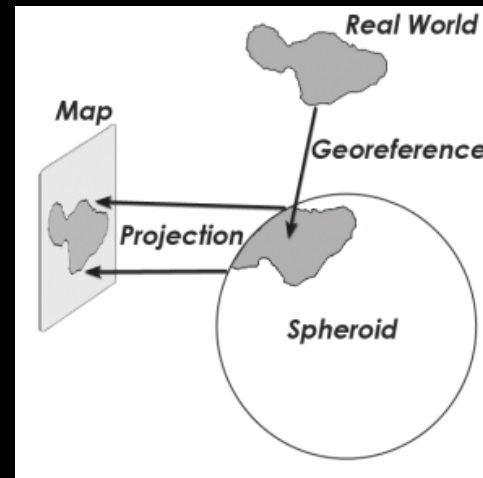
The techniques created by cartographers and mathematicians *to depict all or part of a three-dimensional, roughly spherical surface on two-dimensional, flat surfaces with minimal distortion.*

The need for Map projections

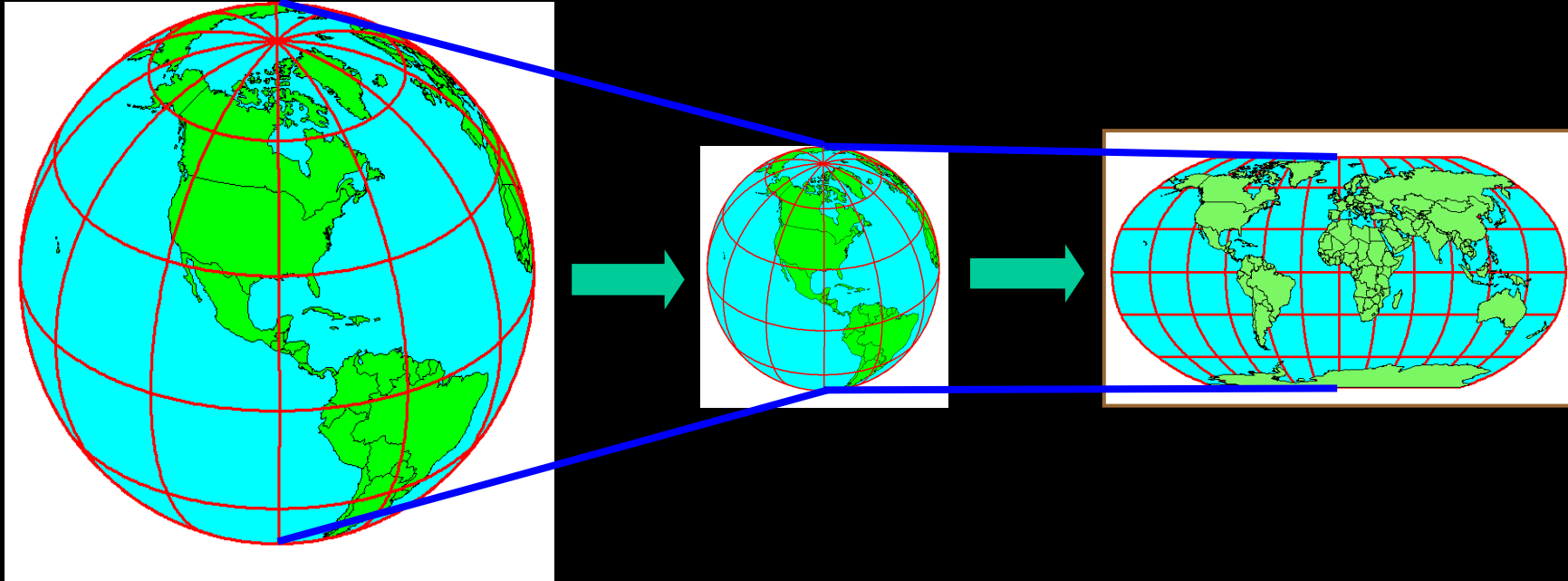
The problem is that:

Earth is 3D

But paper is 2D (flat surface)



Earth to Globe to Map



Earth to Globe:

$$\text{Representative Fraction} = \frac{\text{Globe distance}}{\text{Earth distance}}$$

Globe to Map:

$$\text{Scale Factor} = \frac{\text{Map distance}}{\text{Globe distance}}$$

Map Projections

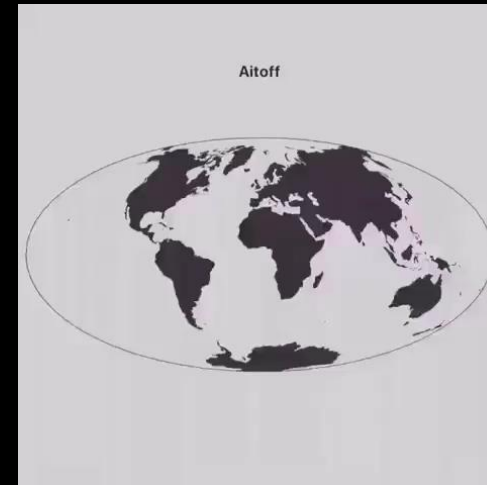
Definition:

A map projection is a mathematical transformation of *locations in the 3-D space of Earth's surface* **onto** the *2-D space of a map sheet*.



This process would certainly **distort** some properties of the real world features, such as **area, shape, distance, and direction**.

However, **there is no universally perfect projection**, and a number of projection systems are in use.



Properties of real world features to be considered for map projection

1) Area

- Is the area measured on the projected map equivalent to that measured on Earth?
- If yes, then the projection is called *equal-area* or an *equivalent projection*.
Examples: Albers Equal Area *Conic*, Lambert *Azimuthal* Equal Area Projections.
- Equal areas achieved by distorting other properties (shape, distance and direction).
- Used for thematic maps (population, land use/land cover, soil type etc).

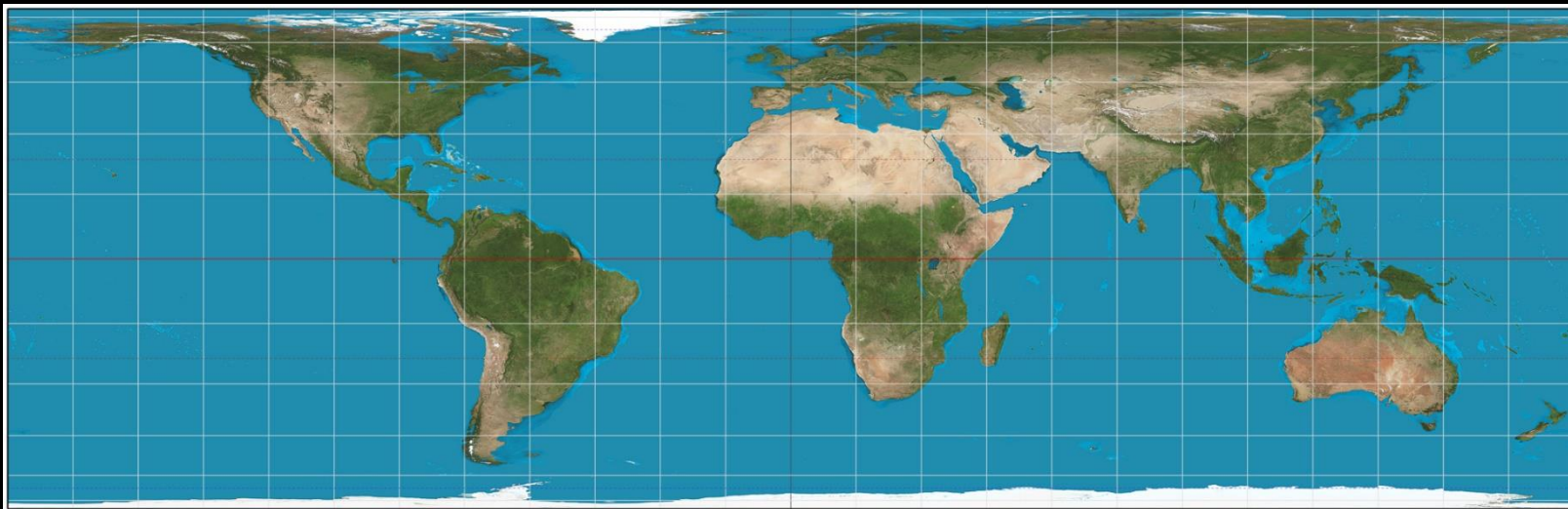


Illustration of Lambert Azimuthal Equal Area Projection

Properties of real world features to be considered for map projection

2) Shape

- Is the shape of a feature on the projected map conforming to its shape on Earth?
- If yes, then the projection is called *conformal projection*.

Examples: Lambert Conformal *Conic*, Mercator Projections.

- Similar shape achieved by maintaining same scale along meridians and parallels in both directions; but it distorts area, distance and direction properties.
- Used for topographic mapping and navigation purposes.



Illustration of (World) Mercator Projection

Properties of real world features to be considered for map projection

3) Distance

- Are distances on the projected map equivalent to that on Earth?
- If yes, then the projection is called *equidistant projection*.

Examples: *Azimuthal* Equidistant, Equidistant *Conic* Projections.

- Distances preserved by projecting standard parallels and central meridians as equally spaced lines on the map; other feature properties distorted.
- Often used in world atlas maps.

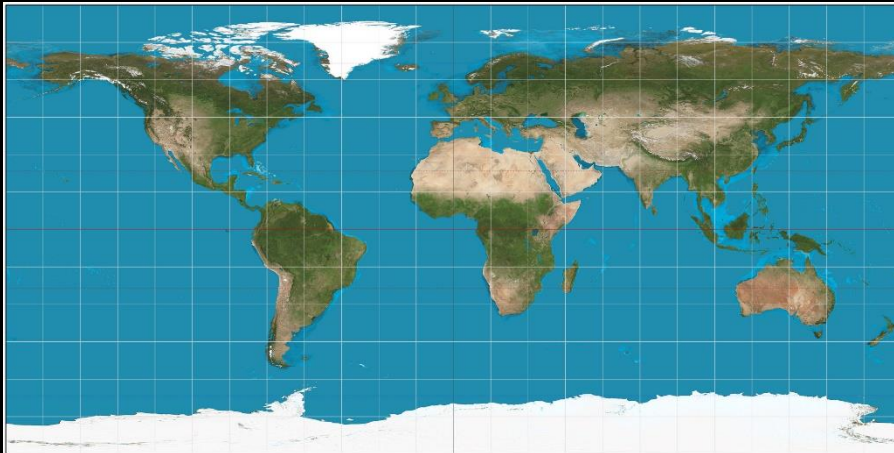
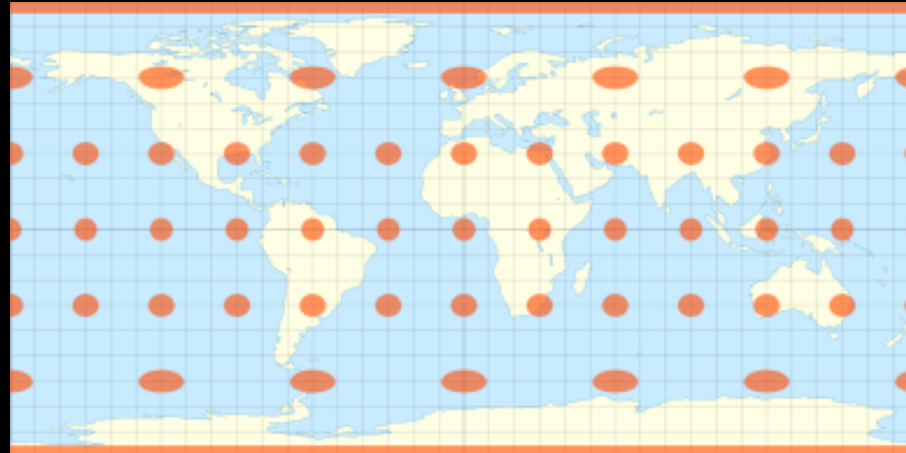


Illustration of Equidistant Projection



Distortions in an equidistant projection

Properties of real world features to be considered for map projection

4) Direction

- Are the directions measured on the projected map same as those on the ground?
- If yes, then the projection is called *true direction projection*.
- Directions can be preserved simultaneously along with other properties also.

Example: Mercator (conformal) projection

- Mainly used for air and sea navigation charts.

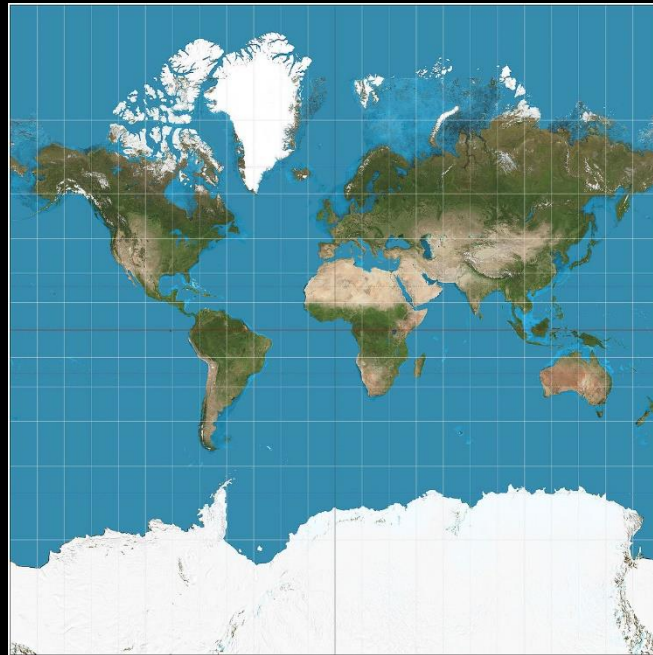


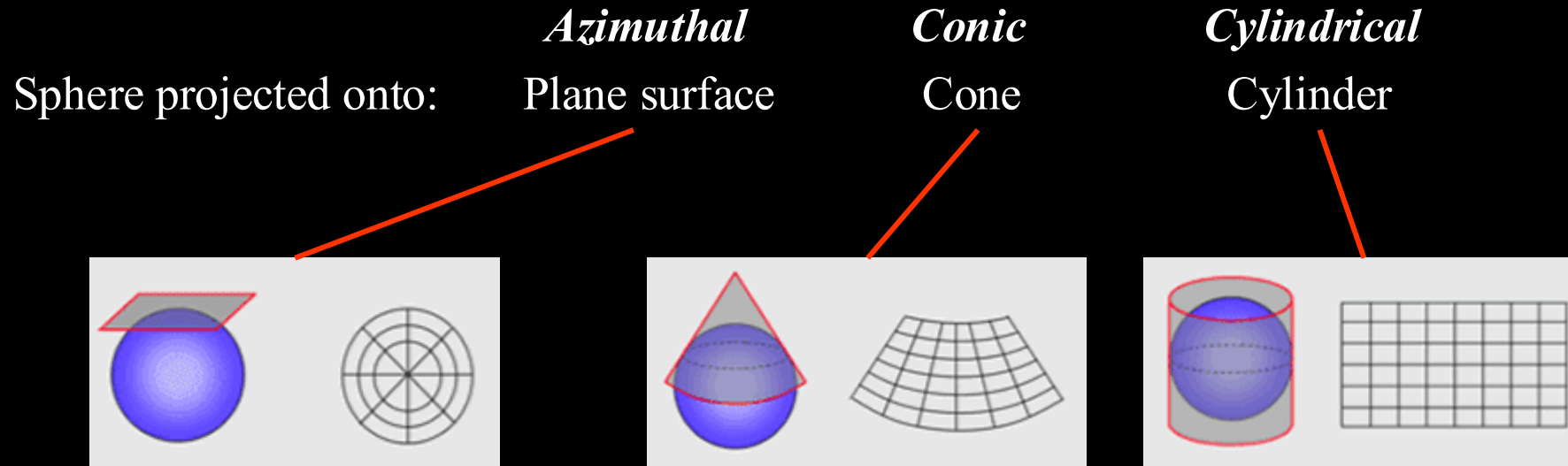
Illustration of (World) Mercator Projection

Classification of Map Projections

1. Based on *property distorted/preserved*:

- **Equivalent:** Preserves area → E.g., Lambert *azimuthal* equal area
- **Conformal:** Preserves shape → E.g., Lambert conformal *conic*
- **Equidistant:** Preserves distance → E.g., Equidistant *conic*
- **True Direction:** Preserves direction → E.g., Mercator *cylindrical*

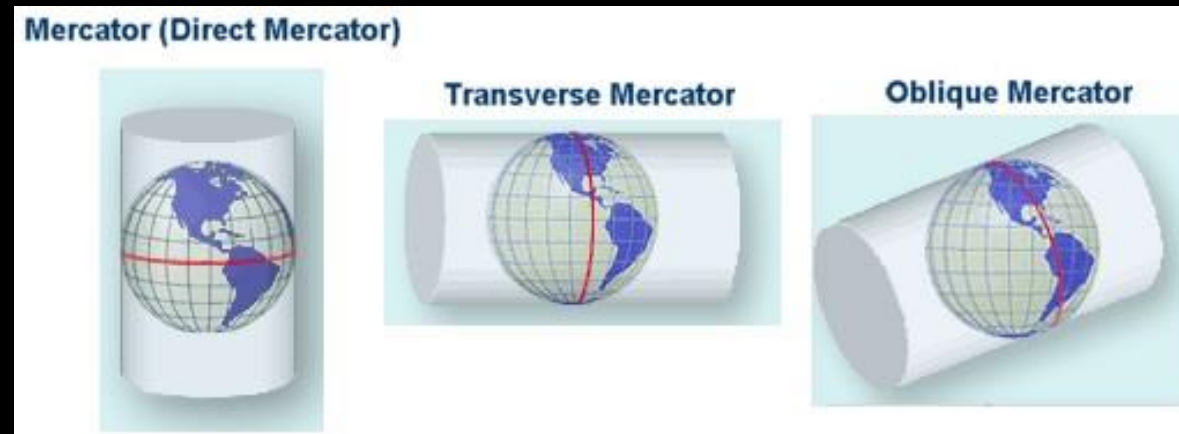
2. Based on *the shape of projection sheet (surface) used*:



Three Basic Types of Map projections (based on projection surface used)

1) Cylindrical:

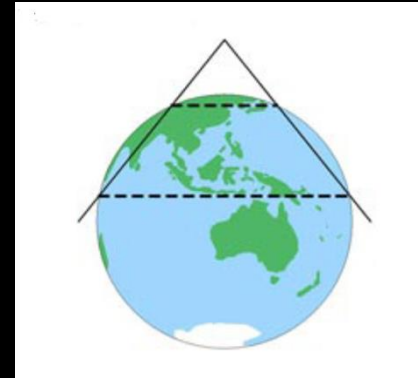
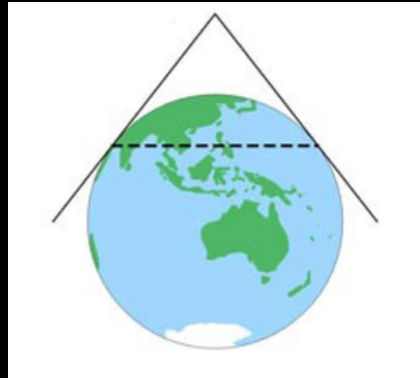
- Analogous to wrapping a **cylinder of paper** around the earth, projecting the earth's feature onto it, and then unwrapping the cylinder.
- Based on the position of the cylinder, there are three main types:
 - i) *Normal/Direct*
 - ii) *Transverse*, and
 - iii) *Oblique*



Three Basic Types of Map projections (based on projection surface used)

2) Conic:

- Analogous to wrapping a sheet of paper around the Earth **in a cone**, projecting the earth's feature onto it, and then unwrapping the cone.
- Based on the cone's intersection with globe, there are mainly two types:
 - i) Tangent (**1** standard parallel)
 - ii) Secant (**2** standard parallels)



Note: Standard Parallels are control latitudes for mathematical transformation, chosen by the cartographer.

- Secant cylinders (with two standard parallels) help in minimizing distortions.

Three Basic Types of Map projections (based on projection surface used)

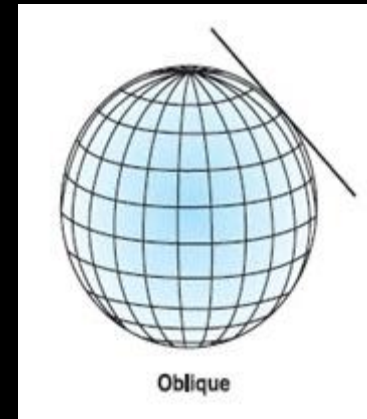
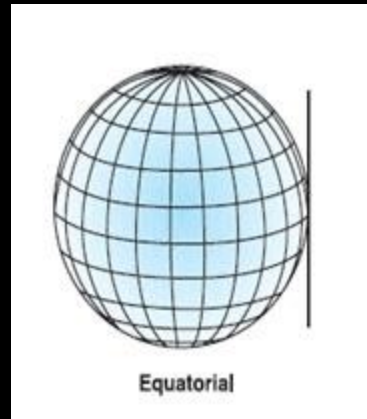
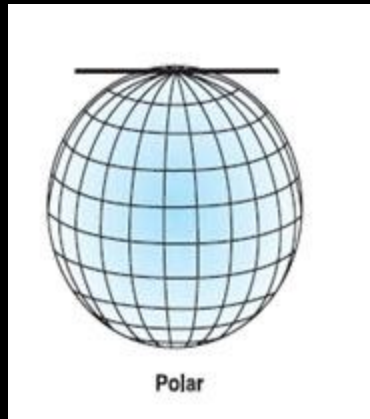
3) Azimuthal:

- Analogous to projecting the positions directly to a **plane**.
- Based on the plane's intersection with globe, there are three main types:

i) Polar

ii) Equatorial, and

iii) Oblique



Types of Map Projections

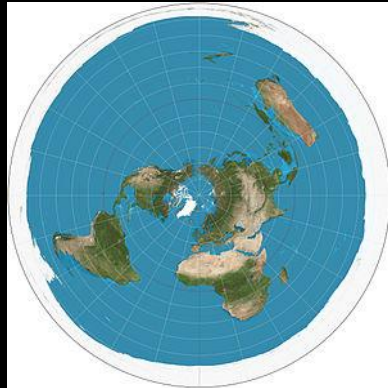
As said earlier, classification can be based on:

1. Property distorted/preserved.
2. Shape of projection sheet (surface) used.

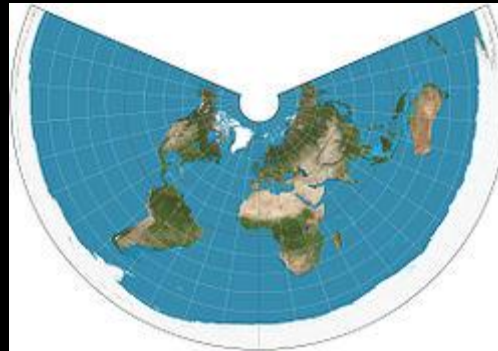
Note:

a) Two projection systems **preserving the same property** can use **different projection surfaces**:

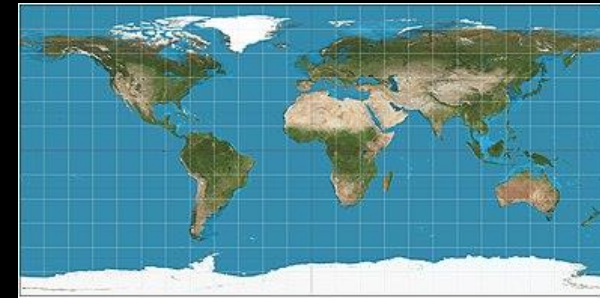
Example: *Azimuth equidistant*



Equidistant Conic



Equidistant Cylindrical



b) Two projection systems **using the same projection surface** can **preserve different properties**:

Example: *Equidistant Conic* (Distance)

Lambert Conformal Conic (Shape)

Hence, based on the mathematical parameters adopted, numerous projection systems are in use.

Commonly used Map Projections

Mercator Projection

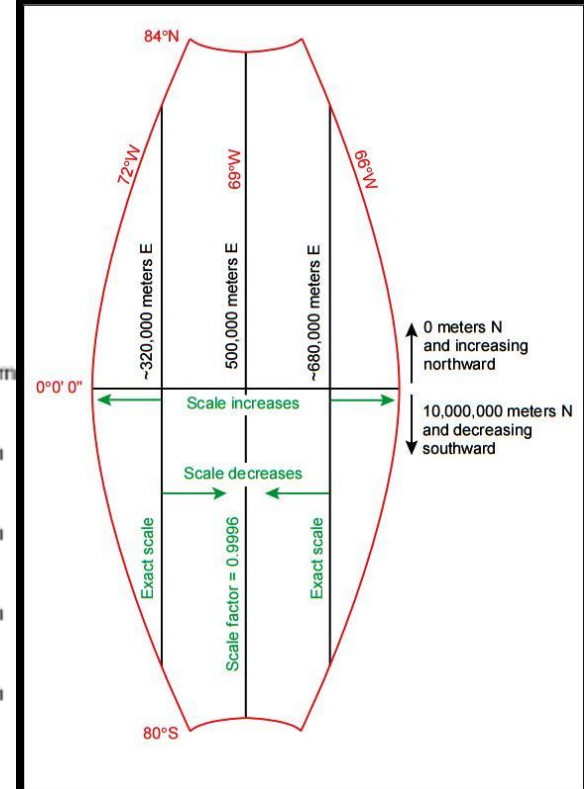
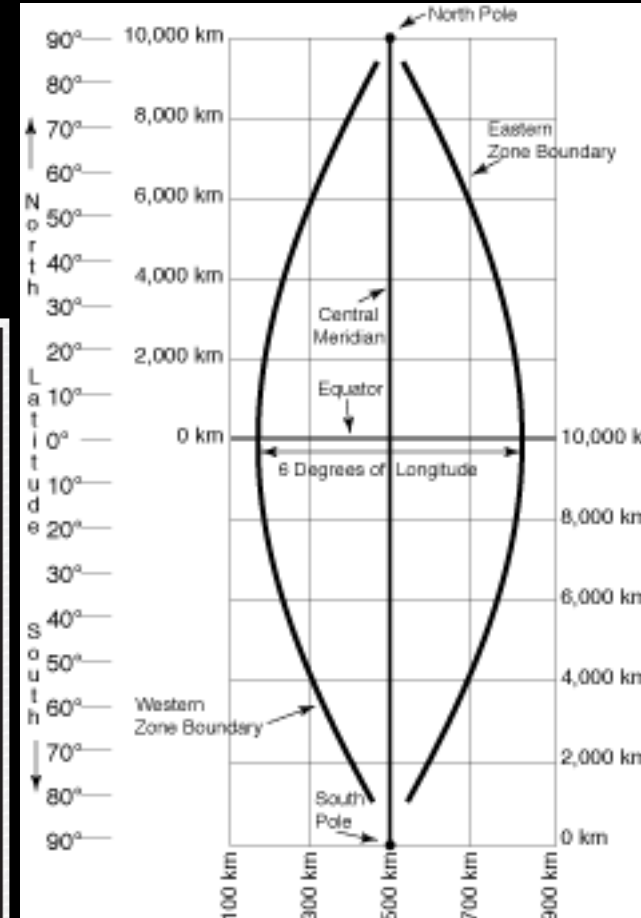
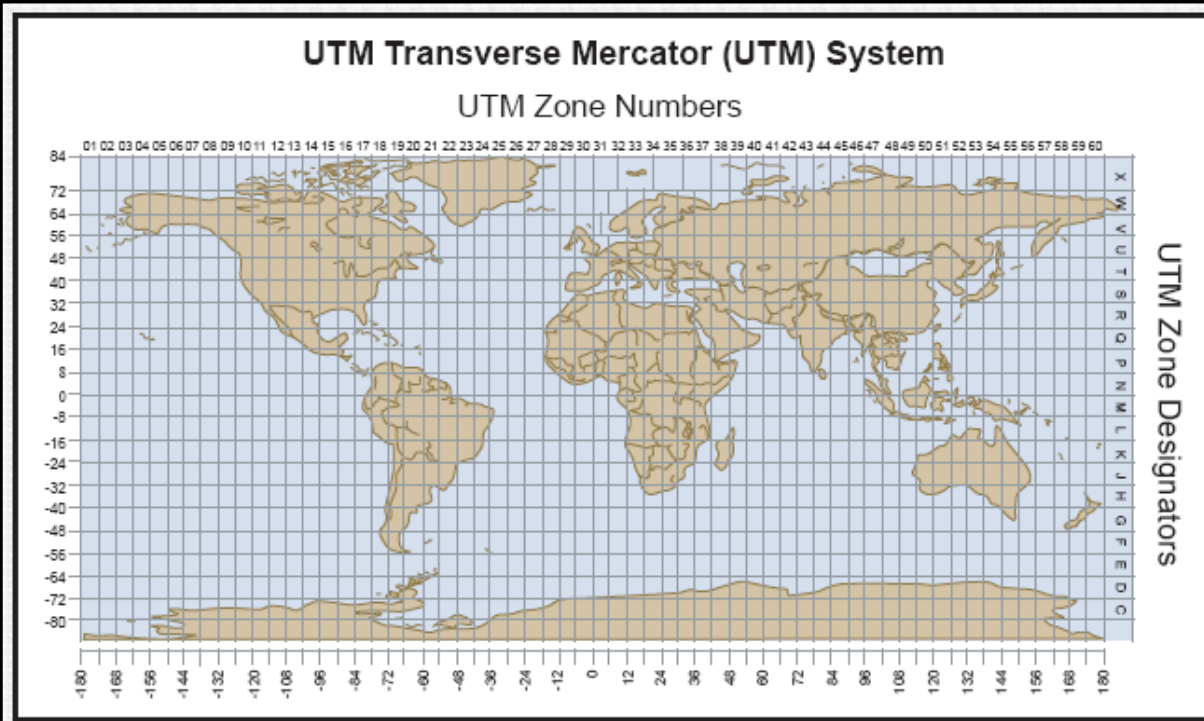
- Cylindrical , conformal projection
- Exaggerations in the high latitudes
- Used in navigation charts and conformal world maps



Commonly used Map Projections

Universal Transverse Mercator Projection

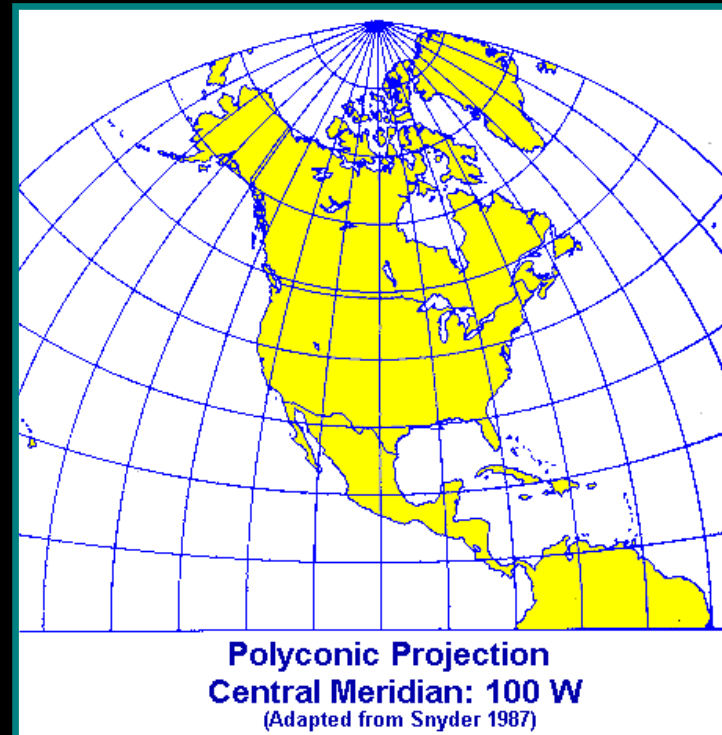
- Cylindrical, conformal projection
- Divides horizontal map surface into sixty zones, each covering 6° longitude.
- It is actually a combination of sixty projections.



Commonly used Map Projections

Polyconic Projection

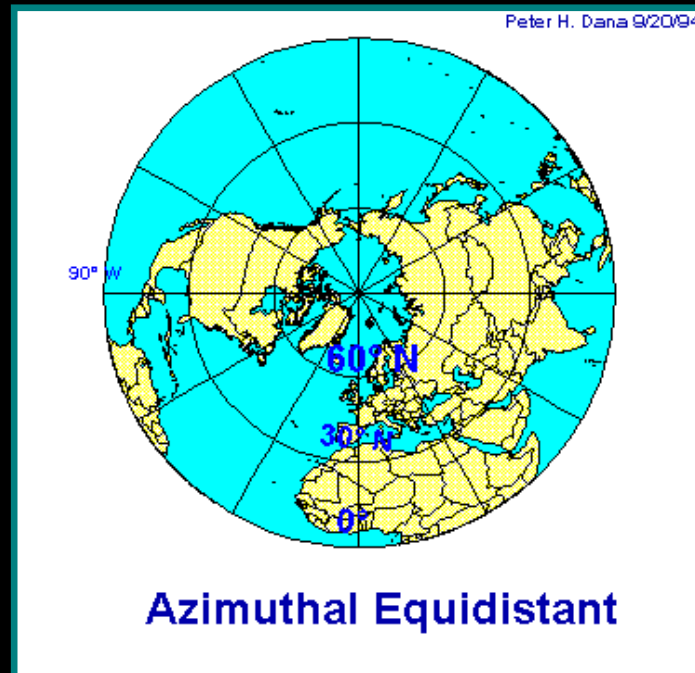
- Conic, equidistant projection
- Scale is true along each parallel and along the central meridian.
- Used for topographic maps; widely used by Survey of India till early 2000s.



Commonly used Map Projections

Azimuthal Equidistant Projection

- Azimuthal (planar), equidistant projection
- Used to show air-route distances.
- Distances from the center are true, distortion radiates out from the center of the map.



Example: UN Logo

Choosing the right Map Projection

Points to Consider (all inter-related):

- Properties to be preserved: Area or Shape or Distance or Direction?
- Purpose of the map:
 - General (atlas)- need to balance shape and area distortions.
 - Specific maps- focus on the property to be preserved.
 - Example: Population density → preserve area;
 - Air-routes → preserve distance
- Scale of the map: Generally, large scale maps (1:10,000 or higher) → Conformal
- Location of the area:
 - Tropical regions → Cylindrical
 - Middle latitudes → Conic
 - Polar regions → Azimuthal
- Shape (extent) of the area:
 - N to S → Cylindrical
 - E to W → Conic
 - Equal extent in all directions → Azimuthal
- Data standards: conform to the national/organizational spatial data standards.

Map Projection and Coordinate Systems in GIS

Map projection → Mathematical transformation of 3-D Earth to 2-D sheet.

Modern GIS packages can handle most if not all the standard projection systems.

Coordinate System → a reference framework that defines the position of features in either 2-D or 3-D space.

GIS mainly use two types of coordinate systems: a) Geographic, b) Projected.

Geographic Coordinate System:

→ Units in decimal degrees, measuring degrees of longitude and degrees of latitude.

→ Use positive values for N and E; negative for S and W.

→ Example: 19.132582 N, 72.916152 E

Projected Coordinate System:

→ Units in linear measurements (x, y).

→ Example: The above location in UTM 43N projection is (280811.3, 2116804.3).

Modern GIS packages are programmed to well handle the conversions between various projection systems or coordinate systems.

Outline

- **Geometric Transformation**
- **Root Mean Square (RMS) Error**
- **Interpretation of RMS Errors on Digitized Maps**
- **Resampling of Pixel Values**

Geometric Transformation

Geometric Transformation is the process of using a set of control points and transformation equations to register a digitized map, satellite image or an aerial photograph onto a projected coordinate system.

Also known as Georeferencing

Types of Geometric Transformation

- **Map to Map Transformation**

Involves transformation of scanned/digitized map into projected coordinates

- **Image to Map Transformation**

Applies to remotely sensed data. The term suggests that the transformation changes the rows and columns of a satellite image into projected coordinates.

A georeferenced image can register spatially with other features or raster layers in a GIS database as long as the datum and coordinate system is the same.

Steps involved in Geometric Transformation /Georeferencing

There are two key steps in Georeferencing,

1. **Assigning** coordinate & projection system to a scanned map/image in physical space by giving known control points using a transformation method.
2. **Re sampling** (filling) of the transformed image/map.

Geometric Transformation Methods

- Different methods have been proposed for transformation from one coordinate system to another.
- Each method is distinguished by the geometric properties it can preserve and by the changes it allows.
- The effect of transformation varies from changes of position and direction, to a uniform change of scale , changes in shape and size.

The following are the transformation methods:

- Equiarea transformation
- Similarity transformation
- Affine transformation
- Projective transformation

Geometric Transformation Methods

The following summarizes these transformation methods and their effect on a rectangular object:

Equiarea transformation-allows rotation of the rectangle and preserves its shape and size.

Similarity transformation- allows rotation of the rectangle and preserves its shape but not its size.

Affine transformation- allows angular distortion of the rectangle but preserves the parallelism of lines(i.e., parallel lines remain as parallel lines)

Projective transformation- allows both angular and length distortions, thus allowing the rectangle to be transformed into a irregular quadrilateral.

General Rules for Geometric Transformation Methods

- **Affine transformation**- for Map to Map or Image to Map transformation if the area is relatively flat (i.e., less undulated terrain)
- **Projective transformation**- for aerial photographs with relief displacement.
- **Polynomial Transformation**- These use surfaces generated from 2nd or higher order polynomial equations to transform satellite images with high degrees of distortion and topographic relief displacement .

The process of general polynomial transformation is commonly called **Warping**.

The **Affine transformation** is also referred to as **1st order polynomial transformation**.

Affine Geometric Transformation

This type of transformation allows rotation, translation, skew and differential scaling on a rectangular object, while preserving the line parallelism.

It can model six kinds of distortion in the remote sensor data including:

- *translation*: Translation shifts its origin to a new location.
- *scale changes*: Differential scaling changes the scale by expanding or reducing in the X and /or Y direction.
- *skew*: allows a non-perpendicularity between the axes, thus changing its shape to a parallelogram with a slanted direction.
- *rotation*: rotates the objects X and Y axes from the origin.

Mathematical Expression of Affine Transformation

Mathematically, the affine transformation is expressed as a pair of 1st order polynomial equations as follows:

$$X' = Ax + By + C$$

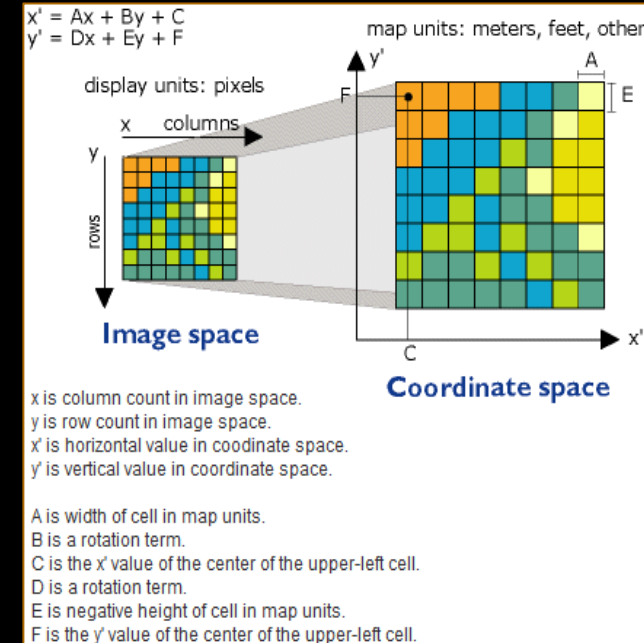
$$Y' = Dx + Ey + F$$

Where

(X', Y') = Output (or Map) coordinates

(x, y) = Input (or image) coordinates

A, B, C, D, E, F = transformation parameters



Goal of the polynomial transformation

The polynomial transformation yields two formulae:

- one for computing the output X' -coordinate for an input (x,y) location, and
- one for computing the Y' -coordinate for an input (x,y) location.

The **goal of the least-squares fitting algorithm** is to derive a general formula that can be applied to all points, usually at the expense of slight movement of the positions of the control points.

Control points #1

Control points play a key role in determining the accuracy of an affine transformation.

For map to map transformation –

It is relatively straight forward task. We need points with known real world coordinates.

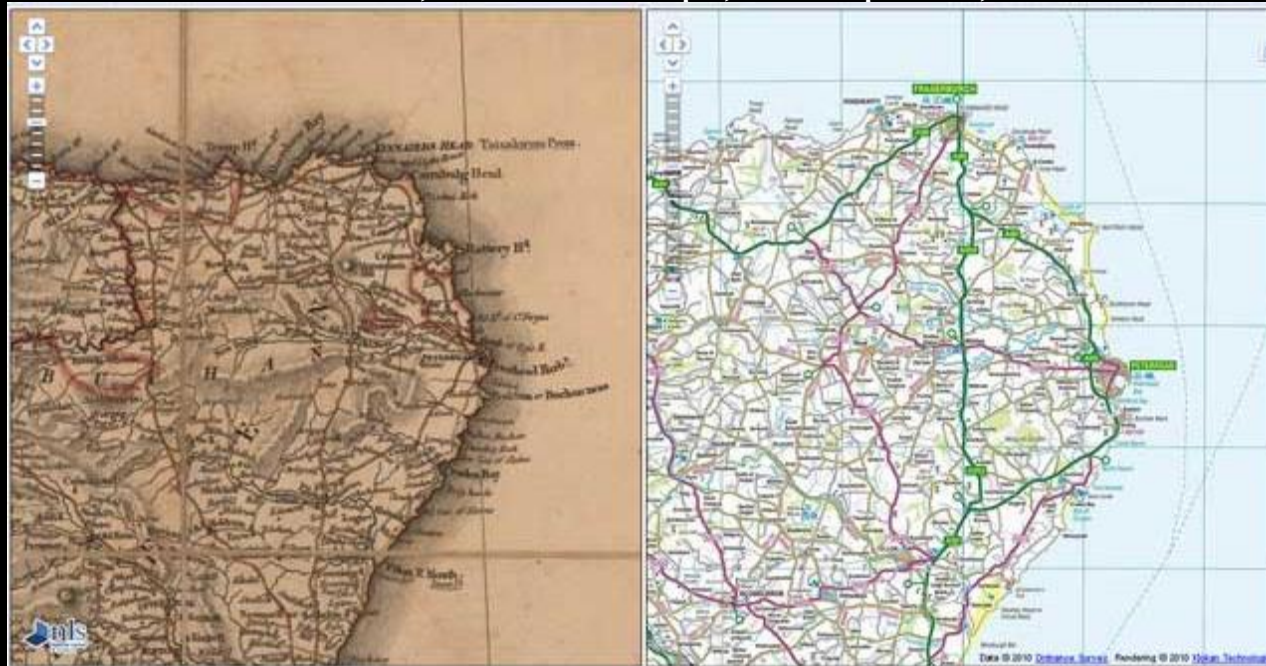
If they are not available we can use points with latitude and longitude values and project them into real world coordinates.

An affine transformation requires a minimum of 3 control points to estimate 6 of its coefficients. But often 4 or more control points are used to reduce problems with measurement errors and to allow of least squares solution.

Control points #2

For Image to map transformation –

- Here it is called as Ground control points (GCP)s.
- GCPs are points where both image coordinates (in columns and rows) and real world coordinates can be identified.
- GCPs are selected directly from a satellite image. Therefore it is not as straight forward as selecting four control points for digitized maps.
- Ideally, GCPs are those features that show up clearly as single distinct pixels and time in variant. **Examples:** Road intersections, Rock outcrops, small ponds, etc.



Assessment of Geometric Transformation (Goodness of the Control points)

Before applying the coefficients to create the rectified output image, it is important to determine how well the six coefficients derived from the least-squares regression of the initial GCPs account for the geometric distortion in the input image.

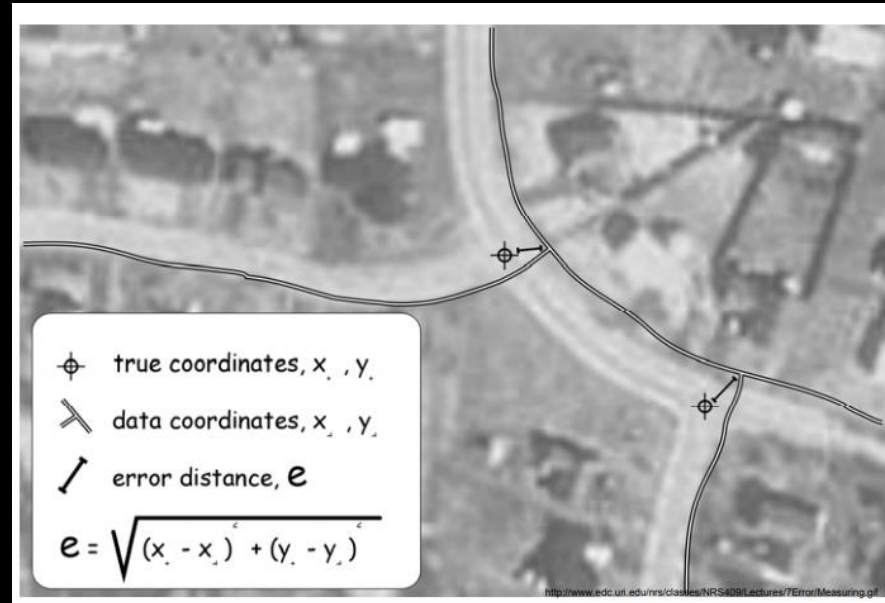
The method used most often involves the computation of the *root-mean-squared error* ($\text{RMS}_{\text{error}}$) for each of the ground control points.

$\text{RMS}_{\text{error}}$ measures the deviation between the actual (true) and the estimated locations of the control points.

Computing the Root-Mean-Squared Error

- Root Mean Squared (RMS) Error is a statistical measure of the error between the calculated coordinates of a GCP and the coordinates of the GCP in the geo-rectified image.
- $$\text{RMS}_{\text{error}} = \sqrt{(x' - x_0)^2 + (y' - y_0)^2}$$

where x_0, y_0 are the original/input coordinates of the GCPs
 x', y' are the computed coordinates of the GCPs
- The average RMS error can be computed by averaging errors from all control points



RMS Error Tolerance

To ensure the accuracy of geometric transformation, the RMS error should be within a tolerance value.

Tolerance value can vary by accuracy and map scale or by the ground resolution of the input data, etc.

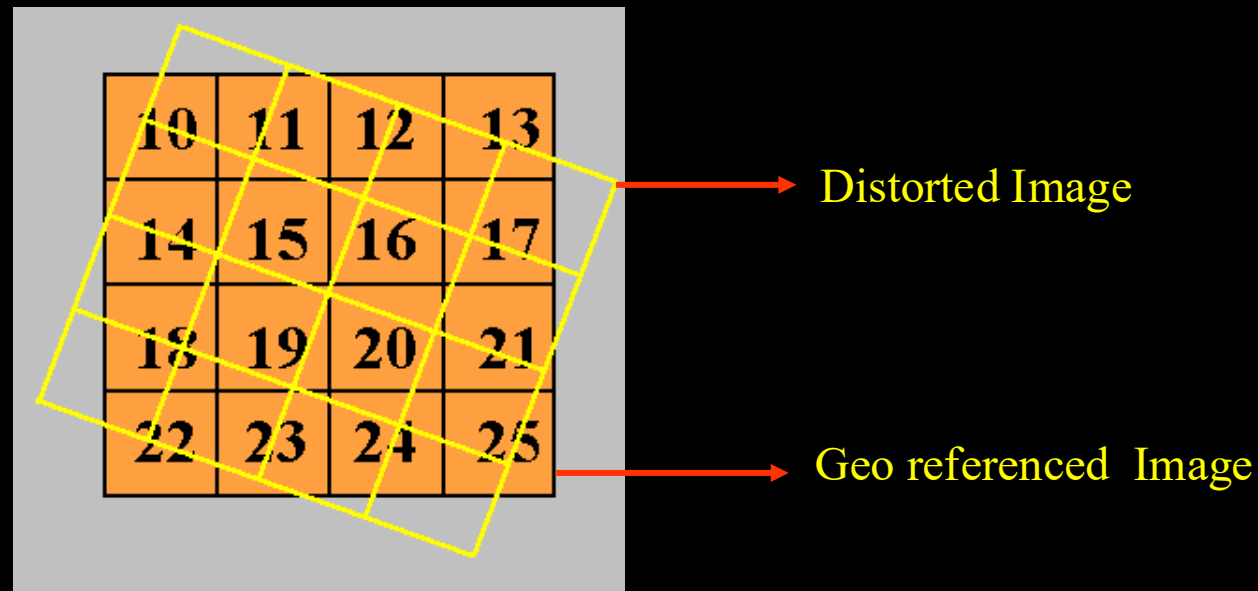
For most data, the tolerance value is specified in the organizational/national spatial data standards.

- For USGS quadrangle maps of 1:24,000 scale, the acceptable RMSE is <6 m.
- For Survey of India (SoI) toposheets of 1:50,000 scale, the acceptable RMSE is <25 m.
- For satellite images, the acceptable limit is: ≤ 1 pixel (input) error in x- and y-directions.

Resampling of Pixel Values (Intensity Interpolation)

Intensity interpolation involves the *extraction of a pixel value from an x, y location* in the original (distorted/digitized) input image/map and its *relocation to the appropriate x', y' coordinate location* in the rectified/transformed/ georeferenced output image.

This *pixel-filling logic* is used to produce the output image line by line, column by column.



Resampling of Pixel Values (Intensity Interpolation)

Most of the time the x and y coordinates to be sampled in the input image are floating point numbers (i.e., they are not integers).

When this occurs, there are several methods of *brightness value (BV)/pixel value intensity interpolation* that can be applied, including:

nearest neighbor,
bilinear interpolation, and
cubic convolution.

The practice is commonly referred to as *resampling*.

Illustration of Common Resampling Methods

- method menu and run the process again
- ☒ repeat with Bicubic as the resampling method

Superimposed portion of rectified output raster. Bold outline indicates current target cell for which a value is being interpolated.

Input Cells Used by Each Resampling Method for the Current Target Cell

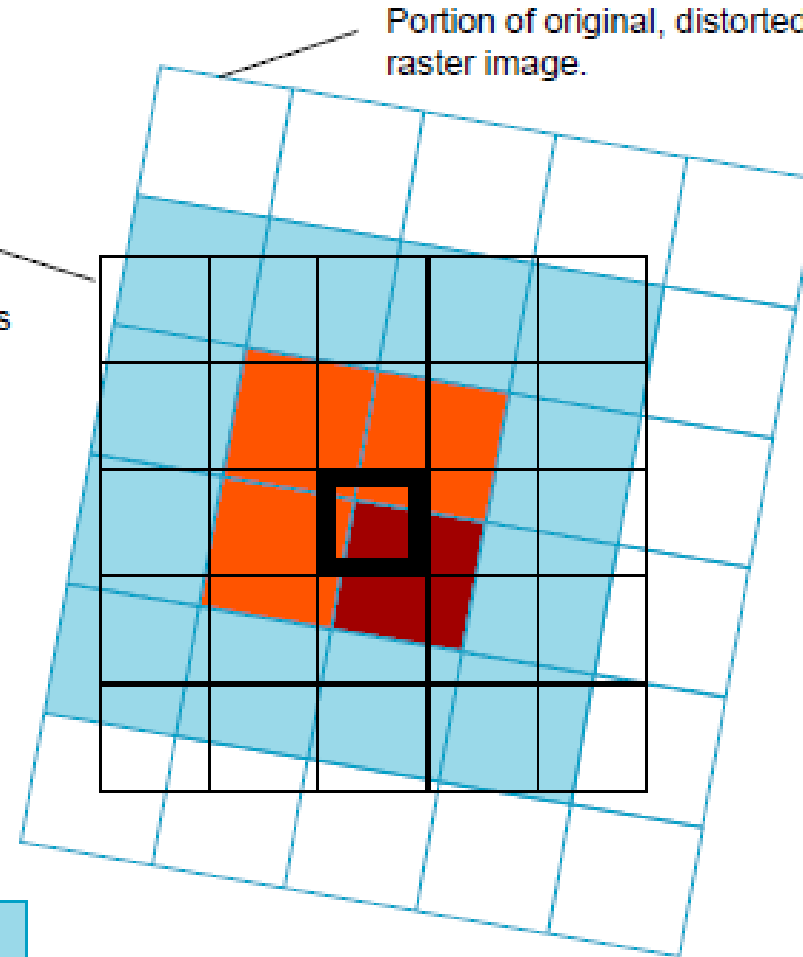
Nearest Neighbor



Bilinear



Bicubic



Overview of Nearest Neighbourhood Resampling

- It uses the digital values from the pixel in the original image which is **nearest to the new pixel location** in the corrected image.
- It offers the advantage of computational *simplicity* and avoids altering the original input pixel values.
- Output matrix may be offset spatially up to one half pixel which cause a disjointed appearance in the output image.

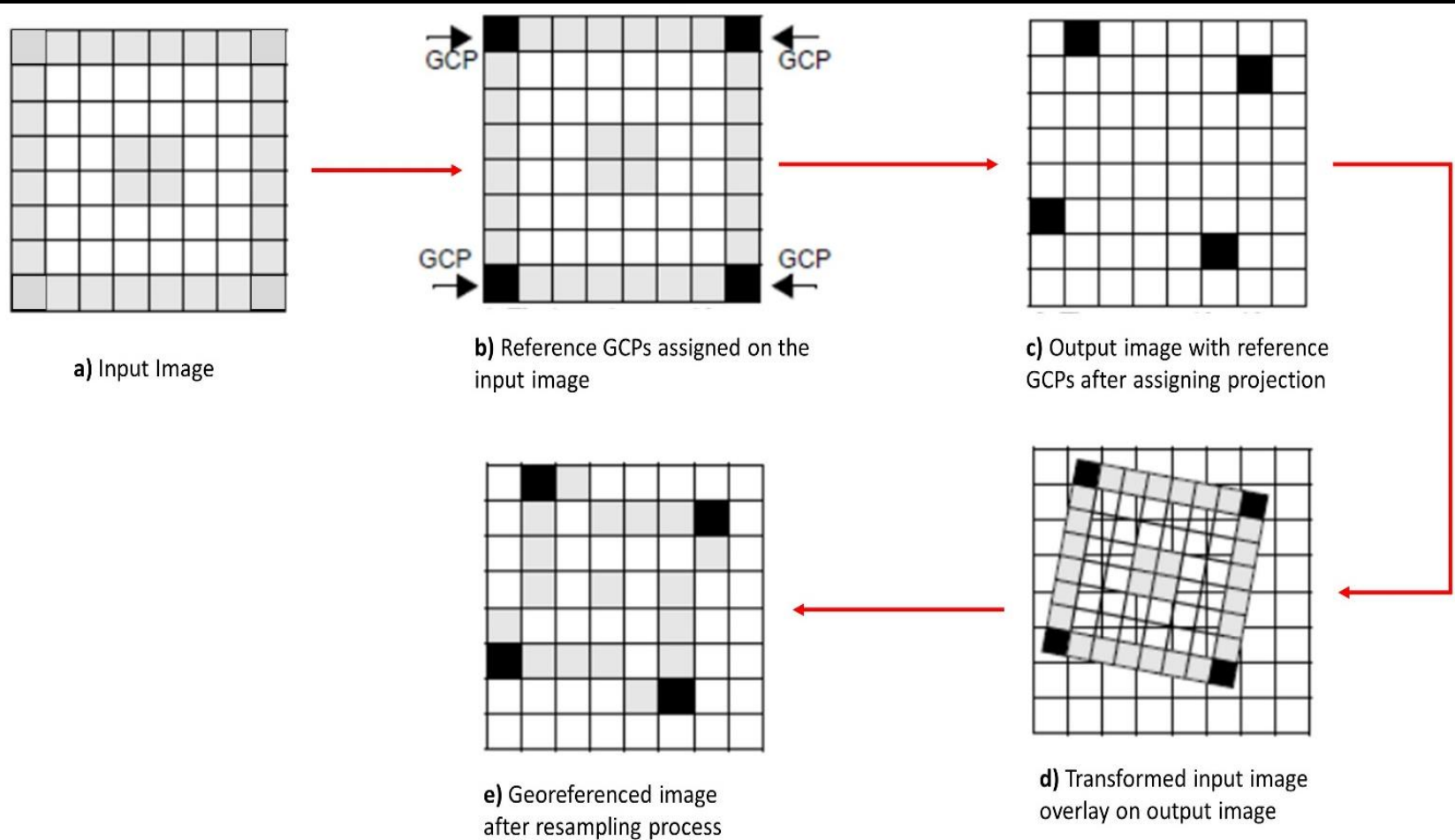
Overview of Bilinear Interpolation

- It takes a distance weighted average of the DN's of the four nearest pixels.
- It is simply a 2D equivalent to linear interpolation.
- It generates smoother appearing re-sampled image.
- This process alters the gray levels of the original image and causes some problem in spectral pattern recognition analysis.

Cubic convolution Resampling

- New pixel values are determined by evaluating the **block of 16 pixels** in the input matrix that surrounds each output pixel.
- It generates smoother appearing resampled image and provides slightly sharper image than bilinear interpolation method.
- This process too alters the gray levels of the original image.

Illustration of a Simple Georeferencing Process



More details can be found at

<https://pro.arcgis.com/en/pro-app/help/main/welcome-to-the-arcgis-pro-app-help.htm>